Assessing L2 sociolinguistic competence: a repetition task approach to /l/ deletion in non-native French

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Abstract

This study sits at the intersection of variationist sociolinguistics and second language acquisition research. It specifically investigates one sociolinguistic variable, /l/ deletion in spoken French, to better understand how non-native speakers acquire native-like (i.e., non-random) use of sociolinguistic variables. Previous studies have shown that advanced non-native speakers do employ /l/ deletion, and, to an extent, that they delete /l/ in meaningfully constrained ways. However, no studies have investigated the relationship between individual speakers' overall deletion rates and their acquisition of native-like deletion patterns. My study thus asks: does a high rate of /l/ deletion necessarily correlate with non-random usage?

To fill this gap, this study draws on a novel experimental paradigm, the Sociolinguistic Repetition Task (Buson et al. 2018; Viana dos Santos et al. 2018). In addition to being more efficient than sociolinguistic interviews, the repetition task has the advantage of testing underlying competence, not performance, as most past studies do.

In a sample of ten advanced non-native French speakers, I find an overall /l/ deletion rate in line with previous studies. Participants were sensitive to native-like constraints on deletion overall, but there was a large degree of individual variation in this regard. Importantly, high deletion rates did not necessarily correlate with good adherence to the native variable grammar. I thus conclude that overall usage of a variable should not be taken as a pure indicator of a speaker's sociolinguistic competence. Instead, my findings suggest the existence of different stages of sociolinguistic competence: first no usage of the variable, then semi-random usage, then, meaningfully constrained usage. These findings paint a more nuanced portrait of the acquisition of L2 sociolinguistic competence and will be an excellent jumping-off point for future research in the field.

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1. Introduction

1.1 Overview

Variationist sociolinguistic research is clearly pertinent to the field of second language acquisition (SLA). Broadly, sociolinguistic SLA research concerns itself with how non-native speakers (NNSs) acquire sociolinguistic variables in their second language (L2). As Bayley and Escalante (2022) observe, "variation is an essential component of language competence" (4). Yet little is known about how NNSs actually acquire native-like patterns of variation.

Fundamental to the present study is the observation that sociolinguistic variation, while random to an extent, is meaningfully conditioned by linguistic and extralinguistic factors. Sociolinguistic competence thus involves not just using a nonstandard variant at a native-like rate but also using the variable in a way that is consistent with native speaker (NS) patterns of usage. That is to say, NNSs must not deploy the variable randomly.

In the service of answering broader questions about the acquisition of sociolinguistic competence, the present study focuses on one sociolinguistic variable: /l/ deletion in the French subject clitics *il* '3.SG.NOM' and *ils* '3.PL.NOM'. An example of /l/ deletion is given in (1b):

(1) a. ils vont à l'école
[**il** vɔ̃ a lekɔl]
3.PL.M go.3.PL to the-school
'They go to school'

b. ils vont à l'école
[**i** vɔ̃ a lekɔl]
3.PL.M go.3.SG to the-school
'They go to school'

Native speakers of Metropolitan French have been found to delete /l/ at a rate of approximately 90% in these pronouns, and deletion is highly conditioned by linguistic factors. For example, /l/ deletion is more common with impersonal uses of *il* than personal uses ((2a) vs. (2b)), and

deletion occurs more frequently before consonants than vowels ((3a) vs. (3b)) (Ashby 1984/2023[1]; Armstrong 1996).

(2) a. il fait beau
[**i** fɛ bo]
EXPL make.3.SG good
'It's nice out'

b. % il fait des crêpes
[**i** fɛ de kʀɛp]
3.SG.M make some crepes
'He is making crepes'

(3) a. ils vont à l'école
[**i** vɔ̃ a lekɔl]
3.PL.M go.3.PL to the-school
'They go to school'

b. %ils achètent des pommes
[**iz** aʃɛt de pɔm]
3.PL.M buy.3.PL some apples
'They buy apples'

SLA literature generally shows that advanced NNSs can approach, but certainly do not meet, native-like rates of use for sociolinguistic variables, consistently over-using stylistically formal variants (i.e., /l/ retention). In the case of /l/ deletion, five studies find /l/ deletion rates between 20–65% during sociolinguistic interviews or role play tasks with advanced NNSs (Thomas 2002; Sax 2003; Nadasdi et al. 2005; Howard et al. 2006; Kennedy Terry 2017). In these studies, "advanced" is taken to mean that the speakers have studied abroad in France and have had several years of classroom French instruction (or, in the case of Nadasdi et al. (2005),

[1] *On Spoken French: An Ashby Reader* reprinted Ashby's major works in 2023. I cite from this book, although the content is identical to the original 1984 article.

are immersion learners). These studies disagree, however, on the extent to which NNSs adhere to native-like linguistic and stylistic constraints on /l/ deletion.

Furthermore, each study finds a high degree of variation in /l/ deletion rates between individual speakers. Generally, more proficient speakers have a higher rate of variable use. However, no study has addressed the correlation between overall /l/ deletion rate and native-like patterns of deletion. My study thus asks: does a high rate of /l/ deletion necessarily correlate with non-random usage? Understanding whether adherence to native-like constraints develop before, after, or in tandem with acquiring a native-like rate of usage will help us better understand the longitudinal development of sociolinguistic competence and more accurately measure discrete stages of competence for both /l/ deletion and other sociolinguistic variables.

A second open question in the existing literature concerns experimental paradigms. In their use of sociolinguistic interviews and role-play tasks, previous studies measure only NNS performance, not competence. Research suggests, however, that under-use of formal variants is partially due to the high processing demands of second language speech (French & Beaulieu 2016). One thus wonders: is it possible that NNSs have underlying knowledge of the NS variable grammar that they are simply unable to express in spontaneous speech? That is to say, what new findings might emerge from testing NNSs' competence independently of their performance?

To this end, my study of /l/ deletion extends a novel experimental paradigm, the Sociolinguistic Repetition Task (SRT), first proposed by Buson et al. (2018) and Viana Dos Santos et al. (2018). In the original SRT, speakers hear a series of formal and informal utterances. Formality is defined in terms of the utterances' semantic content, vocabulary, discourse particles, and the presence/absence of nonstandard phonological and morphosyntactic variants. The innovation of the paradigm is that some of the formal utterances have a single non-standard

variant inserted, and some informal utterances have a single formal variant. After a three-second delay, participants are asked to repeat the utterance back. If they stylistically "homogenize" the utterance by modifying the unexpected variant, this indicates that they associate that variant with a certain indexical value, which is to say that they have mental "schemas" of formal and informal variants. Buson et al. (2018) finds that native French speakers do indeed "repair" stylistically nonhomogeneous utterances in this way, and Viana dos Santos et al. (2018) finds that NNSs of French do so as well, though to a lesser degree than NSs. This finding suggests that NNSs' overuse of standard variants is not due to a lack of awareness of their informal indexical value.

The SRT is a valuable paradigm in that it tests competence, not performance. However, the original SRT only tests awareness of the (in)formality of variants, not the linguistic contexts conditioning their use. I thus designed a new set of SRT tokens which focus specifically on /l/ deletion and which evenly represent all linguistic contexts known to condition the variable. Ten advanced NNSs of French completed my experiment, and in my analysis, I conducted a mixed-effects variable rule (Rbrul) analysis (Johnson 2009) to ascertain whether my participants' correction of nonhomogeneous tokens was conditioned by /l/'s linguistic context. By assessing competence, I hoped that this experimental paradigm would shed more light on the extent to which NNSs non-randomly delete /l/. In eliminating the need to transcribe and code entire sociolinguistic interviews, I also hoped that it would prove an effective and efficient way to compare NNS' behavior across a relatively large corpus in order to begin to understand the effect of individual differences on the acquisition of variable rules.

1.2 Research questions

The specific research questions underlying the present study are:

1. To what extent is adherence to native-like constraints on /l/ deletion correlated with a speaker's French proficiency and overall /l/ deletion rate? Do any highly-deleting speakers use the variable in random ways?
2. Can the SRT be effectively adapted as a paradigm to assess speakers' awareness of linguistic constraints on sociolinguistic variables? If so, what new insights emerge from using the SRT instead of sociolinguistic interviews to test sociolinguistic competence?
3. Overall, how does the /l/ deletion of my speakers contribute to the (limited) existing picture of /l/ deletion in non-native French?

Answers to these questions will contribute to the broader project of understanding the nature of L2 sociolinguistic competence and how such competence is acquired.

1.3 Structure of the present study

The structure of this paper is as follows. Section 2 presents an in-depth literature review around issues in variationist SLA research, /l/ deletion in native and non-native French, and the Sociolinguistic Repetition Task. In Section 3, I present my study's methodology and hypotheses. Section 4 outlines my results, and in Section 5, I discuss the pertinence of these results to larger questions around variationist SLA. Section 6 concludes.

2. Literature review

In this section, I review the pertinence of variationist sociolinguistic research to the field of SLA, highlighting the main research questions that arise at this intersection. I then turn specifically to French sociolinguistic competence, focusing on studies of L1 and L2 /l/ deletion. I argue that there are major gaps in the present literature on L2 /l/ deletion, motivating the present study. I discuss the shortcomings of past studies before presenting the novel experimental paradigm that my study of /l/ deletion adopts.

2.1 Major themes at the intersection of variationist sociolinguistics and SLA

2.1.1 Areas of inquiry

Born of William Labov's pioneering work in the 1960s, variationist sociolinguistics concerns itself with "the fundamental fact that variation in speech occurs in meaningfully systematic ways" (Bayley & Escalante 2022: 4). Variation occurs at all levels of language – lexical, phonological, morphological, morphosyntactic, and pragmatic – and the speaker is continuously faced with a choice between multiple variants to express a single proposition. To take a salient example, the suffix -ing is variably produced as [ɪŋ] and [ɪn] in English. And yet the choice between them is not completely random; variants are associated with different social, stylistic, and linguistic constraints, each of which predict their use. For example, /ɪn/ occurs more frequently on verbs (i.e., running) than nominals (i.e., bedding). It also occurs more in the speech of men than women, is associated with lower socioeconomic status, and occurs more in informal speech than formal speech (Labov 2001). Such quantifiable "variable rules" are a fundamental part of linguistic competence, enabling speakers to assert their identities and speak in contextually-appropriate ways.

It thus should come as no surprise that variable rules would be of interest to the field of SLA, a field fundamentally concerned with the development of communicative competence. Well-formulated by Regan et al. (2009), "This aspect of language acquisition and its related areas is as crucial in enabling learners to communicate with other people as is grammar" (2). That is to say: one of the major challenges of speaking in a second language is simply speaking in a way that NSs are willing to respond to. If one's speech is stiltedly formal or off-puttingly informal, perfect grammaticality can only go so far.

To this end, researchers have applied variationist methodology to SLA since the 1970s (Bayley & Escalante 2022). Early studies focused primarily on L2 English and concerned themselves with what Rehner (2002) retroactively dubbed "Type 1" variation: variation between grammatical and ungrammatical forms in the L1. In the early 2000s, however, researchers turned their attention towards "Type 2" variation: NNSs' systematic use of L1 Sociolinguistic variants (Rehner 2002: 15). The present study inscribes itself in this line of Type 2 L2 variation research.

A major question for those interested in Type 2 variation is simply the frequency with which NNSs employ nonstandard L1 variants. Is L2 speech variable in the same way L1 speech is? Across the board, the answer to this question seems to be: "to some degree." Although advanced NNSs have been shown to employ nonstandard variants across a variety of L2s, they systematically under-employ these variants compared to the NS norm, due in large part to the prescriptive norms enforced in language classrooms (Howard et al. 2013). To return to the example of the variable (ing), Drummond (2012) finds in a study of Polish immigrants to the UK that his non-native English speakers use the informal variant /ɪn/ in 29.1% of cases. Although establishing a definitive "native norm" is a challenge, Tagliamonte's (2012) finding that native speakers of Yorkshire English use the informal variant in roughly 40% of cases provides a useful

point of comparison. Although Drummond's NNSs approach a native-like rate of usage for the variant, they do not meet it.

Clearly, however, frequency alone cannot paint a full picture of sociolinguistic competence. A speaker could, for example, use [ɪn] at a native-like rate but use it predominantly in formal situations, or else predominantly in nominals, which is to say: not in a native-like way. Type 2 research thus also investigates the ability of NNSs to adhere to L1 social, stylistic, and linguistic constraints on variants. To some extent, NNSs do seem capable of acquiring such constraints. In a broader survey of variationist SLA studies, Howard et al. (2013) report generally that "L2 studies across a range of sociolinguistic variants have shown that L2 learners are sensitive to those factors in native-speaker discourse, such that they by no means make random use of a variant irrespective of the linguistic and social context" (347). Howard et al. likewise find that NNSs uphold L1 gender patterns and, to a large extent, formality constraints. However, as I will argue in section 2.2.2, even if NNS' behavior is definitively non-random, these findings should be interpreted with caution. In reality, there is a large gap between random use and truly native-like behavior. Likewise, a very small number of studies have examined NNSs' ability to apply native-like variable rules at all and many studies present contradictory findings for the same variables, even if broad trends point towards an ability to uphold native constraints.

Besides variant frequency and application of variable rules, a third major question that comes with this Type 2 research is: how do aspects of L2 identity condition sociolinguistic competence? Although many such aspects have been studied, including personality, L1 transfer, and learner motivations, two factors have emerged as the most significant: contact with native speakers and overall L2 proficiency. The substantial research around the variable deletion of the

French negative particle *ne,* which Coveney [1996: 55] describes as "possibly the best known sociolinguistic variable in contemporary French," makes for a telling case study of these issues. In what follows, I briefly discuss the literature on *ne* deletion in order to concretize the issues at stake in variationist SLA research.

2.1.2 A case study: influences on *ne* deletion in L2 French

French has a bipartite negative structure: *ne… pas.* The particle *ne* is optional in spoken French, and NSs have been found to delete *ne* at a rate of approximately 80% in Metropolitan French and at almost 100% (i.e., near-categorically) in Canadian French (Coveney 1996; Sankoff & Vincent 1980). I review several studies on NNSs *ne* deletion*,* beginning with the studies that find the least native-like behavior.

Van Compernolle and Williams (2009) report on university students' language use in an online chat platform, presuming that online chats are informal enough to mirror spoken language. They found overall very low rates of *ne* deletion: less than 1% for the first and second-year French students and 20.58% for third-year students. These statistics contrast with a 84.1% deletion rate found in a corpus of online chat by NSs. This data suggests that beginning and intermediate speakers are unable to produce the variant at all, and the university classroom alone is non-conducive to the acquisition of sociolinguistic competence.

Next, Rehner and Mougeon (1999) studied *ne* retention in the informal speech of 40 adolescents in a French immersion school in Ontario, Canada. These students had little exposure to French outside of school. The study's overall finding is that students omit *ne* in 28% of cases, compared to the near categorial deletion reported in native Canadian French. Although this figure is obviously not native-like, it does make an interesting comparison with the even lower rate reported by van Compernolle and Williams (2009). Presumably, although the immersion

students have not been immersed in an authentically Francophone environment, their higher proficiency in French leads to slightly improved sociolinguistic competence. Still, the ultimate conclusion remains that "even the learning of a mildly marked non-standard variant requires a minimum of interactions with native francophones" (Rehner & Mougeon 1999: 22). One might likewise wonder whether L1 exposure simply improves sociolinguistic competence or whether classroom learning actually actively hinders it. Given the prescriptive norms that L2 classrooms reinforce, the latter seems possible; van Compernolle and Williams (2012) report, for example, in a separate study, on learners who "struggled to unlearn the 'invariable' *ne ... pas* structure that they had grown accustomed to using in their previous coursework," even after being explicitly instructed to omit *ne* (200).

By contrast, Regan (1995) analyzed the casual speech of six university students before and after spending a year abroad in France, finding 19% deletion before the year abroad and an impressive rise to 80% immediately after. However, Regan also found that her speakers did not fully abide by the L1 linguistic and stylistic constraints on *ne* deletion. She thus concluded that her speakers had only partially acquired the variable grammar of *ne* deletion, having learned that deletion is a "badge of nativeness," but not quite mastering its usage (Regan 1995: 258).

At the upper end of the spectrum, Deng (2018) interviewed highly proficient NNSs of French living in France, many of whom worked in Francophone workplaces and had Francophone spouses. Deng found an overall *ne* deletion rate of 80.32% and excellent adherence to the native variable rules. Near-native speakers are thus shown to be able to shed the "residual pragmatic conservatism" that shrouds classroom L2 speech and are ultimately able to acquire a variable grammar, at least for this variable (van Compernolle 2015).

The case of *ne* deletion thus highlights the importance of L1 exposure and L2 proficiency to the acquisition of variable grammar. It likewise demonstrates that a native-like deletion rate is not synonymous with native-like patterns of use; both must be considered to have a full picture of a speaker's sociolinguistic competence.

Having established these facts, I now turn my attention to /l/ deletion in French, a variable which has received a far less comprehensive treatment in the literature.

2.2 /l/ deletion

2.2.1 /l/ deletion in L1 French

As one of the "most 'normative' languages in Europe" thanks to the prescriptive influence of the *Académie Française,* mastering prescriptive norms – and when to break them – is an especially pertinent issue to the study of French as a second language (Regan et al. 2009: 15). As one consequence of these prescriptive norms, relatively little work has been done on L1 French sociolinguistic variables to begin with. Ironically, in fact, as with the case of *ne* deletion, L2 usage of the variable has been studied vastly more than the L1 norm. In addition to *ne* deletion, the most commonly studied French sociolinguistic variables are: *nous*/*on* alternation (variable forms of 'we'), *tu/vous* alternation (variable forms of 'you'), and /l/ deletion (Regan 2022). Of the latter three variables, /l/ deletion is the only one to be conditioned by linguistic factors; the *tu/vous* and *nous/on* alternation are purely sociostylistic. /l/ deletion is thus the most interesting candidate for a nuanced study of L2 sociolinguistic competence.

As evoked above, studies of /l/ deletion in native Metropolitan French are limited. To my knowledge, Ashby (1984/2023) and Armstrong (1996) are the only two to comprehensively treat the variable in Metropolitan France since 1980; note the irony that both of these researchers are themselves anglophone.

Ashby (1984/2023) studied /l/ deletion in a corpus of speech recorded in 1976 in Tours, France. As Ashby reports, “The French of Tours is reputed to be among the most ‘pure’; at any rate, Tourangeau French is relatively free of regional particularities” (Ashby 1984/2023: 163). Ashby’s corpus includes 26 speakers, half between the ages of 14–21 and half between 51–64. The corpus is roughly equally split between male and female speakers and represents a range of socioeconomic classes.

/l/ deletion can occur in a variety of contexts: nominative clitic pronouns (*il* (3.SG.M)*, ils* (3.PL.M)*, elle* (3.SG.F)*,* and *elles* (3.PL.F)), articles (*la*, *les*, *leur,* lui), and words ending in *-ble*, *-cle*, and *-ple* (/kapab/ for *capable* ‘capable’). Overall, however, Ashby found that /l/ deletion occurs the most frequently in the subject clitic pronouns *il* (3.SG.M) and *ils* (3.PL.M). Note that *il* can also have an impersonal (3.SG) meaning, as in *il fait beau* (‘it is nice out’). Accordingly, most L2 studies of the variable focus specifically on these two pronouns.

Overall, Ashby found a deletion rate of 88% for impersonal *il,* 88% for the plural *ils,* and 63% for personal *il.* He likewise found that deletion is significantly more common before consonants and the yod glide (as in *il y a* (‘there is/there are’)) than vowels or pauses. In his analysis, Ashby used the Varbrul 2 program, which calculates the probability of each factor affecting /l/ deletion. One represents the highest possible probability, and zero the lowest. His findings are reproduced in Table 2.1 and Table 2.2 below:

Table 2.1: Ashby (1984/2023)'s findings for /l/ deletion by clitic

Clitic	Observed Frequency	Relative frequency (% of /l/ deletion)	Varbrul 2 probability
il (impersonal)	1544/1755	.88	.745
ils	621/706	.88	.725
il	549/874	.63	.635

Table 2.2: Ashby (1984/2023)'s findings for /l/ deletion by phonetic environment

Environment	Observed Frequency	Relative frequency (% of /l/ deletion)	Varbrul 2 probability
__C	1414/1663	.85	.790
__yod	1078/1190	.90	.733
__pause[2]	12/24	.50	.495
__V	337/857	.39	.090

In terms of demographic factors, Ashby (1984/2023) found that males delete /l/ at a slightly higher rate than females, that younger speakers delete significantly more than the older speakers, and that upper-middle class speakers delete somewhat less than the lower socioeconomic groups. From this, he concluded that /l/ deletion in the subject clitics is a linguistic change in progress but is not necessarily correlated with specific social demographics.

The second major study on /l/ deletion, Armstrong (1996), is the first to consider speech style as a factor influencing the variable. Armstrong's corpus was recorded in 1990 in north-eastern France and is reported to be relatively free of regionalisms. Participants were

[2] Ashby notes: "'Pause' includes hesitation phenomena (*euh* 'uh', *ah* 'oh', etc.) and various noise interferences." (Ashby 2023: 169)

11–18 years old and were recorded in an "interview" style (one on-one with the researcher) and a "conversation" style (participants conversing with their peers without the researcher present). Social class was not reported. Like Ashby, Armstrong observed /l/ deletion in a variety of contexts; here I report only his findings for the subject clitic pronouns *il* and *ils*.

Armstrong reports "very nearly categorical" /l/ deletion with the impersonal *il,* especially in *il y a* ('there is/there are') and *il faut* + infinitive ('one must + infinitive'). He also reports that /l/ is elided "categorically" in the plural *ils* (Armstrong 1996: 11). For the personal *il,* he reports an overall deletion rate of 98.4% when preceding a consonant and only 12.4% pre-vocalically. He also finds a large effect for style shift, with higher deletion rates in the informal style. These results are summarized in Table 2.3. Armstrong also finds slightly higher deletion rates among the 16–19-year-olds in his corpus than those aged 11–12. Finally, like Ashby, he finds that females delete /l/ slightly less than males.

Table 2.3: Summary of Armstrong (1996)'s findings

Environment	Interview (more formal)		Conversation (less formal)		Both styles	
	N	Deletion %	N	Deletion %	N	Deletion %
Impersonal *il*	Near categorical					
ils	Categorical					
Personal *il* + C	355	97.5%	473	99.1%	828	98.4%
Personal *il* + V	352	7.1%	441	16.5%	793	12.4%

Armstrong does not opine directly as to why his deletion rates are so much higher than Ashby's. However, he does refer often to the 'non-standard' speech of his informants in both the interview and conversational styles. Armstrong broadly defines his use of non-standard:

"lower-class and/or casual language;" perhaps, then, the lower deletion rate is attributable to the overall lower register of the corpus. This would indicate that /l/ deletion is strongly conditioned by speech style. Likewise, it is possible that the age of his informants is a major factor; as Armstrong suggests, perhaps /l/ deletion in the subject pronouns is not a linguistic change in progress, as proposed by Ashby, but is in fact age-graded. This conclusion is supported by the observation that /l/ deletion has been observed in *il* since the 12th century; there is no obvious explanation for why the variable should be spreading so rapidly only now (Ashby 1984/2023: 164). Finally, it cannot be ruled out that the variable is, in fact, regionally-conditioned.

From these findings, it is difficult to establish a definitive native speaker norm. Research in more modern corpora is certainly called for, especially given the possibility that /l/ deletion is a rapidly-accelerating linguistic change in progress. However, from these two studies, the following conclusions do seem well-supported:

1. /l/ deletion is relatively more common in *ils* and impersonal *il* than personal *il*
2. /l/ deletion occurs at an overall rate of at least 80% in *ils* and impersonal *il* and at least 60% in personal *il*
3. /l/ deletion is strongly disfavored prevocalically

I take these facts about /l/ deletion to be the native speaker norm in this paper. Having established L1/l/ deletion patterns, I now review previous studies of the variable in L2 French.

2.2.2 /l/ deletion in L2 French

Although it has not been treated as comprehensively as *ne* deletion, /l/ deletion has been examined in a handful of studies. Most of these studies focus specifically on deletion in the subject clitic pronouns; the present study follows suit. I briefly review each study in chronological order before synthesizing their findings.

Thomas (2002) studied /l/ deletion in a corpus of Canadian university students before and after they spent a year studying abroad in France. His speech data came from sociolinguistic interviews and a stylistically neutral reading passage. The study specifically focuses on /l/ deletion in pre-consonantal *il.* Thomas found an overall deletion rate of 28.7% with virtually no difference between personal *il,* impersonal *il,* and *ils,* a deviation from the NS norm. He also found that his informants deleted /l/ slightly more after the year abroad than before and slightly more than a control group who had not studied abroad.

Sax (2003) studied /l/ deletion as part of a larger Ph.D dissertation. Sax's speech data was collected via formal and informal role plays among three groups of American university students who had studied abroad for various lengths of time: 0–2 weeks, 2–5.25 months, and 8.5 months–4 years. Sax found a 4% deletion rate for the first group, 24% for the second, and 62% deletion for those who had spent more than 8.5 months abroad. A Varbrul (variable rules) analysis accordingly found time abroad to be the strongest predictor of /l/ deletion in the data.

Sax also conducted Varbrul analyses for following phonological segment, formality, and /il/ type (singular vs. plural, impersonal vs. personal, and *il y a* vs. other). She found following phonological segment to be a strong predictor of deletion, with *il* + V strongly disfavoring deletion (2% deletion; factor weight of 0.06), *il* + /j/ slightly favoring deletion (45%; factor weight of (0.57), and *il* + C strongly favoring deletion (48%; factor weight of 0.75). Formality mattered to a small extent, with the informal context slightly favoring deletion. *Il y a* likewise slightly favored deletion over non-*il ya a* tokens, plural tokens slightly favored deletion over singular, and impersonal tokens slightly favored deletion over personal. This ordering of constraints appears to mirror the NS variable grammar.

Nadasdi et al. (2005) conducted sociolinguistic interviews with high school French immersion students in Toronto, Canada. The students did not speak French at home and had never had extended contact with NSs of French. The researchers found a deletion rate of only 2% in the pronouns *il* and *ils,* suggesting that, despite the students' high proficiency, their immersion education had been especially non-conducive to the acquisition of this sociolinguistic variable. Given the low deletion rate, there is no analysis of factors conditioning deletion in the study.

Howard et al. (2006) studied /l/ deletion in subject pronouns among Irish university students specializing in French. Four of their speakers had never been to France and fifteen had spent a full academic year abroad. All had 5–6 years of French study prior to university. Howard et al. found an overall deletion rate of 6% for speakers who had not studied abroad and 36% for those who had.[3] In a GoldVarb 2001 (variable rule) analysis, the researchers found that token type was the most important factor in predicting deletion, with impersonal *il* favoring deletion (45%; factor weight of 0.632), followed by *ils/elles* (29%; factor weight of 0.520) and personal *il* (20%; factor weight of 0.353). Following phonological segment was also found to condition deletion, with *il* + V slightly disfavoring deletion (34%; factor weight of 0.459) and *il* + C slightly favoring deletion 32%; factor weight of 0.546). These patterns are mostly in line with NS constraint rankings, although the factor weight for *il* + V is higher than expected, and the factor weight for *ils* is lower than expected. Likewise, the study finds no significant effect for style. The study notably also finds a high rate of individual variation among speakers: from 5% to 48% /l/ deletion among those who had studied abroad.

French and Beaulieu (2016) studied whether explicit instruction of sociolinguistic variables, including /l/ deletion, leads to more native-like performance. As a second innovation, they also studied /l/ deletion in two different task types: planned and unplanned oral production.

[3] These numbers also include /l/ deletion in *elles*

Participants were advanced NNSs of French at a Canadian university. None had lived in a French-speaking environment, though the university was in a partially French-speaking city, and all participants had had some degree of contact with L1 French speakers in their everyday lives.

In the planned speech task, participants were asked to write a short dialogue. Then, after five minutes of preparation, they read the dialogue out loud. In the unplanned speech task, participants were asked to tell a brief narrative based on a series of images. On the planned production task, speakers deleted /l/ at a rate of 6.2% before the period of instruction and 48.1% after their classroom instruction. Unplanned production was not tested prior to instruction, and participants deleted at a rate of 8.3% in spontaneous speech after instruction. This 8.3% consists entirely of tokens of *il y a;* deletion was not observed in any other context. From these results, the researchers concluded that the cognitive demands of spontaneous second language speech are a large factor in NNSs' overuse of formal variants. Their ability to delete /l/ in planned, but not spontaneous speech reflects a performance error, not a lack of declarative knowledge.

Finally, Kennedy Terry (2017) studied /l/ deletion in sociolinguistic interviews with American learners of French studying abroad in France. Kennedy Terry found an overall /l/ deletion rate of 20.5% in the pronouns *il* and *ils.* An Rbrul (variable rule) analysis found token type to be an important factor in deletion, with *ils* strongly predicting deletion (31.9%; factor weight 0.80), followed by impersonal *il* (23.7% ; factor weight 0.65). Personal *il* disfavored deletion (6.9%; factor weight 0.38). Phonologically, *il* + C slightly favored /l/ deletion (17.3%; factor weight 0.55), and *il* + V slightly disfavored deletion (14.2%; factor weight 0.46).[4] These findings generally align with NS patterns, although deletion in *il* + V is unexpectedly high. Time abroad was also found to be a statistically significant predictor of /l/ deletion, as was social integration with NSs while abroad.

[4] These findings include deletion in *elle* and *elles*

Like Howard et al., Kennedy Terry also found a large range of individual deletion rates in her corpus: from 0% to 68.2%. Although an Rbrul analysis was not performed to analyze whether higher deletion rates correlate with better adherence to native-like deletion patterns, the researcher did find that higher deletion rates do not necessarily correlate with the emergence of deletion in different /il/ types. For example, the speaker who deleted at 68.2% only showed deletion in impersonal *il* and *ils,* but a speaker who deleted at 21.6% showed deletion in all four categories (impersonal *il, ils,* personal *il,* and *elle(s)*). Overall, however, the acquisition of deletion in different token types consistently followed the pattern: impersonal *il >> ils >>* personal *il >> elle/elles.* That is to say, the speakers who deleted in only one category all deleted in impersonal *il,* and no speaker deleted in personal *il* without also deleting in *ils* and impersonal *il.* Kennedy Terry does not offer a clear interpretation of these results. However, I take them to suggest that adherence to native-like factor weights may in fact correlate with higher overall deletion rates. Although deleting in more categories than the 68.2% speaker, the 21.6% speaker may well have been deleting randomly, whereas the 68.2% speaker's marked preference for impersonal *il* and *ils* shows good sensitivity to the NS pattern. Rbrul analyses of individual speakers' performances are clearly warranted, as is a comparison with the effect of phonological context.

A summary of all studies reviewed is presented in Table 2.4. Several key trends emerge. First, study abroad, and specifically interaction with NSs while abroad, is necessary to acquire the variable at all. With study abroad, individual speakers' deletion rates can surpass 50%, but none truly approach native-like rates of /l/ deletion. Without study abroad, deletion rates stay below 6%.

Table 2.4: Summary of previous L2 /l/ deletion studies

Study	Participant demographics	Overall deletion %	Conforms to L1 norm?		
			Effect of /il/ type	Effect of phonetic context	Effect of style
Thomas (2002)	Canadian university students post-study abroad	22.7%	No	N/A	N/A
Sax (2003)	American university students with and without study abroad	4% without study abroad; 62% with study abroad	Yes	Yes	Yes
Nadasi et al. (2005)	Canadian high school immersion students	2%	N/A	N/A	N/A
Howard et al. (2009)	Irish university students with and without study abroad	6% without study abroad; 36% with study abroad	Partially	Partially	No
French and Beaulieu (2016)	Canadian university students without study abroad after explicit instruction of the variable	48.1% in planned speech; 8.3% in spontaneous speech	N/A	N/A	N/A
Kennedy Terry (2017)	Americans studying abroad in France	20.5%	Yes	Partially	N/A

As a second trend, speakers seem to approximate native-like patterns of deletion, but to varying degrees of success. Sax (2003) is the only study to find truly native-like factor weights containing deletion; the other studies show that NNSs' deletion is not as systematically constrained as NSs', or else, in the case of Thomas (2002), they have actually acquired the wrong pattern. The acquisition of variable rules thus warrants further research.

As a third trend, speakers exhibit high rates of individual variation where individual performance is studied. Kennedy Terry (2017) is, however, the only study to analyze individual adherence to NS factor weights, and her brief discussion leaves many questions unanswered, particularly the question of whether variable rules are acquired before, after, or simultaneously with the acquisition of native-like deletion rates.

Finally, French & Beaulieu (2016) make a very convincing case for the effect of task type on deletion rates. Their study raises the question: has any previous study actually successfully tested what speakers know about /l/ deletion? Or simply how they are able to perform? A separation of competence and performance – especially as it relates to the acquisition of native-like factor weights – is called for, and it is precisely this gap that my study attempts to fill. To do so, I employ an innovative experimental paradigm, the Sociolinguistic Repetition Task (SRT), first introduced in Buson et al. (2018) and later employed by Viana dos Santos (2018; 2021). In what follows, I describe the paradigm before turning to its applications for the present study.

2.3 The Sociolinguistic Repetition Task (SRT)

The SRT was first proposed by Buson et al. (2018) as a way to evaluate whether speakers have coherent cognitive schemas of sociolinguistic variants. In the original experiment, native French speakers heard 24 utterances, of which half were stylistically formal (using formal discourse participles, elevated vocabulary, and standard phonetic and morphosyntactic variables, such as *ne* and /l/ realization), and half were stylistically informal. Half of these tokens were then made "nonhomogeneous" by incorporating a single stylistically informal variant into the formal sentences or one stylistically formal variant into the informal sentences.

During the experiment, participants heard the utterance spoken aloud. They then saw a single digit number appear on a screen for three seconds, then were asked to repeat the number they had just seen, followed by the utterance they had just heard. During analysis, responses were either coded as "identical repetition" or "non-identical repetition;" i.e., participants heard a nonstandard variant and repeated it as standard. The researchers predicted that participants would tend to correct nonhomogeneous utterances to be homogeneous based on Kiesling's (2011) observation that "the numerous linguistic choices with which an individual is confronted when producing an utterance (heterogeneity) form predictable inter- or intra-individual patterns (order) at the collective level, insofar as they are correlated with social or linguistic factors" (as cited in Buson et al. 2018: 804). This tendency to homogenize utterances is indeed what they found, suggesting that speakers have coherent mental schemas of sociolinguistic variables. The researchers conclude: "the sociolinguistic repetition task in combination with methods from the field of social cognition (e.g., social priming) opens up a promising direction for studying the cognitive representation of sociolinguistic varieties" (Buson et al. 2018: 814).

Soon after, Viana Dos Santos et al. (2018; 2021) adapted the SRT to an SLA context. The studies explicitly challenge the notion of incomplete mental schemas as the reason that NNSs overuse formal sociolinguistic variants. Indeed, in an experiment with 66 intermediate to advanced learners of French, Viana dos Santos finds that NNSs "correct" stylistically nonhomogeneous sentences at a statistically significant rate: 87.5% towards formal homogeneity, and 54.4% towards informal homogeneity. Although these rates are lower than Buson et al.'s NSs, the findings nevertheless suggest that NNSs do have better sociolinguistic competence than they are able to demonstrate during sociolinguistic interviews.

It remains to be seen, however, whether NNSs show sensitivity to variables' linguistic constraints in this context or whether the mental schemas are limited to stylistic information. Likewise, the original SRT included only 24 tokens representing a variety of sociophonetic and morphosyntactic variables. Participants' competence around individual variables could thus not be tested. In the following section, I describe how I adapt the SRT to test specifically for competence around linguistic constraints on /l/ deletion.

3. Methodology

In this section, I describe the methodology for my experiment. I discuss my participants, the construction and procedure of my modified SRT, and my data analysis protocol. I then present my study's hypotheses.

3.1 Participants

3.1.1 Recruitment

10 participants were identified after responding to an advertisement seeking non-native speakers of French who had spent time living or studying abroad in a Francophone country and who spoke French at a self-identified "advanced" level. One additional speaker participated in a pilot version of the experiment. This speaker had an advanced level of French but had not spent time abroad; their data was not formally analyzed.

Of the 10 non-pilot participants, four were recruited at a study abroad center in France, three from the researcher's home university, and two through the researcher's personal connections. Participants' language proficiency was not evaluated by the researcher before they began the experiment.

3.1.2 Demographics

3.1.2.1 The Bilingual Language Profile (BLP)

Participants responded to a modified version of the Bilingual Language Profile (BLP), a survey that assesses language dominance in bilinguals (Birdsong et al. 2012). The BLP was selected for the experiment because it has been widely validated, is easy to use, and produces a single, quantified "score" for various aspects of each participant's language background and competence. Note, however, that all BLP questions are "self-report"; participants' language proficiency was never assessed objectively.

In addition to the survey's standard questions on language background, language proficiency, and language attitudes, additional questions were added to assess participants' mode of learning for French and English (at home, at school, in an immersion environment, etc.), and to what degree they feel comfortable speaking in a "native-like way" in both English and French. A copy of the survey questions can be found in Appendix B.

3.1.2.2 Biographic Information

Of the 10 participants, eight were undergraduate students, and two had received an undergraduate degree in the past year. On an open-ended question, seven reported their gender as female, and three reported their gender as male. All speakers reported being fully comfortable speaking English at the time of the experiment; six participants had learned English since birth and four began learning English between the ages of three and five. Further information about these four speakers' L1s was not collected.

In terms of language background, eight participants were classroom learners of French, one had learned while spending part of their childhood in France, and one had learned at an immersion school in the United States. Participants began learning French between the ages of five and 19 (median 11) and had a median of eight years of classes taught in French. Besides the participant who had spent four years living in France, nobody had spent more than one year in a Francophone country. No participants regularly spoke French at home.

3.1.2.3 Language proficiency, attitudes, and use

The language proficiency section of the BLP asks participants to rank how well they read, write, speak, and understand French on a scale of zero ("Not well at all") to six ("Very well"). Zero thus represents the lowest possible composite score, whereas 24 represents perfect self-perceived proficiency. Participants' scores in this section ranged from nine to 22 with a mean score of 15.2

and a median of 15.[5] Individuals' composite scores are reported in Table 3.1; scores for individual questions can be found in Appendix C.

Of particular interest to this experiment, the language attitudes section of the BLP asks participants to self-rate for the questions "I feel capable of speaking French in a native-like way," "It is important to me to use (or eventually use) French like a native speaker," and "I want others to think I am a native speaker of French." The scale was again zero ("disagree") to six ("agree"); the composite scores for these three questions ranged from six to 18 with a mean of 12.6 and a median of 13.5. Participants' scores for each question are reported in Appendix C and their composite attitudes score is reported in Table 3.1.

Finally, the BLP's language use section asks what percentage of time participants use French with friends, with family, at school/work, when talking to themselves, and when counting. Composite scores in this section are computed between zero and 50. Participants' scores ranged from one to 13 with a mean of 4.6 and a median of three, which is to say, none of my participants regularly use French outside of the classroom. The composite scores are reported in Table 3.1 and full results are in Appendix C.

[5] I acknowledge that such low self-reported proficiency scores seem at odds with the fact that all speakers also self-identified as "advanced" speakers of French when signing up for the study. One potential explanation could be that participants evaluated their abilities in comparison to other *advanced speakers,* not objectively.

Table 3.1: BLP language scores

Participant	Composite Proficiency Score (0-24)	Composite attitudes score (0-18)	Composite use score (0-50)
A	12	11	1
B	13	16	2
C	9	11	5
D[6]	20	15	1
E	19	14	6
F	9	8	1
G	13	6	2
H	17	13	4
I	18	18	13
J	22	14	11
Mean	15.2	12.6	4.6

3.2 Overall study protocol

Data was collected in a pilot with one speaker, then over four sessions on three dates. Data from the pilot experiment was used to refine the SRT tokens and is not included in analysis.

Participants A, B, C, D, and E had their audio data collected via the built-in microphone on a Sony ICD-PX370 mono digital voice recorder at a sampling frequency of 44,100 Hz. Participants A, B, and C completed the experiment in a sound-resistant booth at a study abroad center in France, Participant E completed all the tasks in a quiet bedroom, and participant D completed the experiment in an office of the study abroad center. Other people were in the room

[6] Participant D did not respond to one question on the BLP, however they responded with 5 for the other three questions. For the sake of a complete composite score, I thus decided to also use a score of 5 for the fourth score.

wearing headphones while Participant D read the reading passage and completed the repetition task, but this did not affect the quality of the recordings.

Participants F, G, H, I, and J completed the experiment in a completely sound-proofed recording studio. For these participants, recordings were collected on a high-quality audio system at a sample rate of 48,000 Hz. Although this setting was much less naturalistic than the others, I judged that the benefits of obtaining such high-quality acoustic data for potential future research outweighed the potential negative effects of conducting the experiment in a formal-feeling environment. Given that no acoustic analysis was performed in the present experiment, however, the type of recording device (Sony or recording studio) is irrelevant.

Participants were told that they were completing an experiment on second language speech but were not given any more specific details. They were first briefed on the study protocol, and they provided their informed consent. Next, they completed the BLP on a laptop computer. Participants then completed the two experimental tasks: a reading passage and the SRT.

The reading passage was included in the experiment to have a guaranteed source of /il/ tokens in case SRT data proved unusable for any reason. The passage was approximately 250 words long and was taken from the transcript of a Radio France podcast about different sports vocabulary in the Francophone world. It was selected for the experiment because of its stylistic neutrality and frequent use of the pronoun /il/ in various contexts. Participants were invited to look over the passage before reading, then were asked to “Please read it out loud, imagining it as an informal conversation between friends.” The researcher remained in the room during the reading. Data from the reading passage was not ultimately analyzed in this paper, however the

data may prove valuable to future research. After reading the passage out loud, participants were given instructions for the SRT.

3.3 SRT methodology

3.3.1 Procedure

With the exception of the number of tokens, this portion of the experiment followed the protocol from Buson et al. (2018) and Viana dos Santos (2021) as closely as possible. Participants sat in front of a laptop computer with a microphone either beside the laptop or approximately one foot in front of them. The experiment ran on PsychoPy (Peirce et al. 2019). Participants heard recorded utterances, then a single-digit number appeared on the screen for three seconds. After the number disappeared, participants were prompted to state the number they had just seen, followed by the utterance that they had just heard. As outlined by Buson et al. (2018: 808), the three-second pause forced participants to partially reconstruct the utterances, making use of their sociolinguistic schemas instead of pulling the utterances directly from auditory memory.

Participants were instructed to press the spacebar when they were ready to hear the next utterance. Utterances were presented in a random order, and the single-digit numbers were likewise random. The experimenter remained in the room for two practice tokens, then participants were left alone to complete 56 experimental tokens. The experiment was self-paced, and participants were encouraged to take their time. Participants were likewise reassured that their comprehension of the utterances would not be tested and that it was expected that they might not understand or be able to recall every word. They were asked to simply repeat the utterances to the best of their ability. Most participants took just over twenty minutes to complete the task, including the practice trials.

3.3.2 Experimental tokens

Broadly speaking, tokens for the experiment were short utterances containing *il* or *ils* and which fell under four categories: formal homogenous, formal nonhomogeneous, informal homogeneous, and informal nonhomogeneous. “Formal” and “informal” refer to the stylistic register of the overall utterance. Formal utterances used elevated vocabulary, formal discourse particles, included formal sociostylistic features such as *ne, dont,* and optional liaison, and mainly had content related to societal debates.[7] Informal utterances used a lower register and often included slang, used informal discourse particles, included sociostylistic features such as *ne* deletion, *ce que* (‘that which’) for *dont* and non-realized optional liaison, and discussed everyday situations. These components are summarized, with examples, in Table 3.2.

[7] *Ne* is a negative particle. *Dont* is a relative pronoun.

Table 3.2: Components of formal vs. informal utterances[8]

Distinctively formal and informal components are given in bold

Formal	Informal
Elevated vocabulary, including rare words - Example: *Il a accepté d'adopter un **amendement polémique***	Casual vocabulary, including slang - Example: ***Ouais** il **bosse** tout le temps, c'est **abuser**.*
Ne-retention - Example: *Il **ne** s'agit que d'une première épreuve pour le budget de l'Etat*	Ne-deletion - Example*: euh non, ce **ø** sont pas des danseurs. Ils font la fête dans un club quoi*
Liaison - Example: ***C'esT un** pays accueillant où il fera certes bon vivre.*	Non-realized liaison - Example: ***C'est un** sale gosse, je te jure, il est grave*
Relative with *dont* - Example: *Je m'interroge sur la façon **dont** ils conçoivent l'enseignement en général*	Relative with *que* - Example: *C'est quand même un peu chelou **ce que** je t'ai parlé tout à l'heure*
Formal discourse particles - Examples: *oui bon, euh non*	Informal discourse particles - Examples: *oh mais, ben, quoi, hein*

In keeping with the analysis that /l/ deletion is an indexically informal variable, homogeneous tokens are defined as informal tokens with /l/ deletion in *il(s)* and formal tokens with /l/ retention. Nonhomogeneous tokens are informal tokens with /l/ retention and formal tokens with /l/ deletion. Examples from each category are given in Table 3.3; a full list of tokens is given in Appendix D.

[8] English translations of these utterances are given in Appendix D.

Table 3.3: Example Utterances[9]

	Formal	Informal
Homogenous	Oui bon, **iL** est vrai que les prix Nobel tendent à récompenser des travaux d'avant-garde	Ouais **i'** bosse tout le temps, c'est abuser.
Nonhomogeneous	Pour subvenir à ses besoins, bien entendu, **i'** faut d'autres sources de revenus.	C'est un appart avec un pote **iL** y a pas de télé

As outlined in Section 2.3, the major innovation of this experiment is that it also controlled for the linguistic context of *il(s)* in the experimental utterances. Given the factors known to condition /l/ deletion in native speech, this experiment controlled for the following variables: plurality (*il* vs. *ils*), following segment (consonant vs. vowel), and personality (personal vs. impersonal). *Il y a* was also treated as its own category because of its status as a fixed expression.

The 28 experimental utterances contained 31 instances of *il(s)*, as three utterances contained two instances of the pronouns. Of the utterances, fourteen were formal and fourteen were informal. Of these, four in each category were homogenous and ten in each category were nonhomogeneous. Although it would have been ideal to have equal numbers of homogeneous and nonhomogeneous tokens, it was deemed overall more important to limit the number of tokens such that the experiment could be completed in approximately twenty minutes. Given the need for statistically-rigorous data on whether participants altered the nonhomogeneous tokens to be homogeneous and the hypothesis that participants would not modify homogeneous tokens to become nonhomogeneous, I thus decided to skew the distribution of tokens in this way.

[9] English translations of these utterances are given in Appendix D.

On the other hand, tokens were distributed over the three linguistic variables as evenly as possible. The total number of tokens for each variable is presented in Table 3.4 below. The total numbers add up to 31, not 28, per variable, as each instance of *il(s)* in utterances with more than one pronoun was counted separately. Overall, the configuration of variables yielded seven distinct categories, with either four or five tokens per category. This distribution is further elaborated in Table 3.5.

Table 3.4: Number of tokens per linguistic variable

Context	Number of tokens
Singular	16
Plural	10
Precedes consonant	13
Precedes vowel	13
Personal	18
Impersonal (excluding *il y a*)	8
Il y a	5

In addition to the 28 experimental utterances, the experiment included 28 filler utterances which did not use the pronoun *il(s)*. Each filler did, however, include at least one of the sociostylistic variables detailed in Table 3.2. By manipulating these variables, I produced fourteen homogeneous fillers and fourteen nonhomogeneous fillers, each set evenly split between formal and informal. The filler tokens are also given in Appendix D.

Table 3.5: Distribution of tokens across categories

Context	Formal H	Formal NH	Informal H	Informal NH	Total
Personal *il* + C	0	2	1	1	4
Personal *il*+ V	1	1	0	2	4
Personal *ils* +C	1	1	0	3	5
Personal *ils* + V	1	1	0	3	5
Impersonal *il* + C[10]	0	2	1	1	4
Impersonal *il* + V[11]	1	1	0	2	4
Il y a	0	2	1	2	5
Total	4	10	3	14	31

When possible, utterances were taken directly or lightly adapted from Appendix A of Viana dos Santos (2021). This procedure yielded a total of 23 utterances. The remaining 35 experimental utterances were constructed by the researcher. To ensure authenticity, I adapted formal tokens from sentences in articles on LeMonde.Fr, a major French newspaper. I adapted informal tokens from French TV shows and lifestyle vlogs on Youtube. In both the experimental and filler utterances, explicitly political and polemical topics were avoided, as was any vulgar or potentially offensive language.

The utterances for the experiment were recorded by a male native French speaker in a completely soundproof recording studio. The speaker recently completed an undergraduate degree in Paris, and I, as a non-native French speaker, determined his speech to be free of any noticeable regional accent. Participants F, G, H, and I had met the speaker and were able to identify the voice as his in the experiment.

[10] Excluding *il y a*
[11] Excluding *il y a*

Following the pilot experiment, I determined that some tokens were too long or complex to be repeated successfully. Although it was not always possible to shorten the recorded utterances, I did splice the audio clips to eliminate some filler words and non-essential clauses where possible.

3.4 Analysis protocol

Data was analyzed solely by the researcher. I listened to each speaker's recording from the SRT and classified each instance of *il(s)* as '/l/ deleted,' '/l/ retained,' or '/l/ ambiguous.' Although there were 310 tokens for analysis, in many cases, participants could not recall the utterance or else modified it to delete the /l/. Likewise, in other cases, the speaker modified the utterance such as to change the linguistic context for /il/ (i.e., making a singular *il* plural or replacing a following consonant with a vowel). All of these tokens, totalling 92, were removed. Of the 218 remaining for analysis, 56 were initially judged /l/ ambiguous.

After this first pass of analysis, I listened to the 56 ambiguous tokens a second time, this time also examining the corresponding spectrogram in Praat (Boersma & Weenink 2023). I first examined the spectrograms of a random set of tokens with clear /l/ retention and deletion. I observed that, when /l/ was clearly present, the spectrogram showed a visible dip in F2 and/or rise in F1 at the end of /i/. When /l/ was clearly deleted, F1 and F2 stayed constant through the entire vowel.

When re-examining the ambiguous tokens, in some instances, the /l/ became classifiable upon the second listen alone. In other cases, the spectrogram showed a clear dip in F2 or rise in F1, which suggested the presence of an /l/. If this formant data aligned with my auditory impressions, the token could be classified. Finally, in eight cases, the token remained truly

ambiguous both auditorily and on the spectrogram. These eight tokens were thrown out, leaving a total corpus of 206 tokens across the 10 speakers.

Each speaker in my corpus was assigned a pseudonym (letters A–J). Each speaker's instances of /l/ retention or deletion were treated as an individual token for statistical analysis. Further details on the Rbrul (variable rule) analysis undertaken are given in section 4.3.2.

3.5 Hypotheses

Based on the /l/ deletion literature and my study protocol, I made the following hypotheses:

1. Although native speakers delete /l/ at around 80%, I predicted that my speakers would delete /l/ in 20-40% of cases, which would echo the findings of Thomas (2002), Howard et al. (2009), and Kennedy Terry (2017). In line with the findings from previous studies, I also predicted that speakers with higher BLP proficiency scores would be more likely to delete /l/ overall.
2. I hypothesized that, because my study tested competence instead of performance, my speakers would show a high sensitivity to native-like factor weights. I specifically predicted that *il y a*, *ils*, and *il* + C would be highly significant predictors of deletion, and that personal *il* and *il* + V would strongly disfavor deletion.
3. Given that /l/ is known to be a stylistically informal variable, I predicted that informal utterances would favor deletion over formal utterances. However, I predicted that the (non)homogeneity of a token would not influence deletion.
4. I predicted that the presence or absence of /l/ in the experimental token would not significantly affect participants' deletion. This would indicate that participants do have their own mental schemas for the variable and are not reliant on cues from native speakers.

5. Given the limited findings from Kennedy Terry (2017), I predicted that better adherence to native-like factor weights would correlate with both a higher overall deletion rate and higher self-reported proficiency.

4. Results

In this section, I present the key findings from my experiment. I begin by discussing task completion, overall /l/ deletion, and the effects of linguistic factors on deletion at the corpus level. I then turn my attention to individual speakers' sensitivity to linguistic factors.

4.1 Task Completion

As described in Section 3.4, one challenge of the experiment was that many sentences proved too long or complicated for participants to successfully repeat. This was true of both filler tokens and experimental tokens. Of the 310 possible tokens, 210 were ultimately deemed usable. The remainder were thrown out because /il/ was deleted by the participant, because the linguistic context for *il(s)* was changed by the participant, or because it was impossible to tell whether /l/ was present. Presumably, the failure to repeat tokens reflects a lack of language proficiency; Buson et al. (2018)'s NSs only failed to repeat the utterance in 4.7% of cases as opposed to my 32.2%. Curiously, Viana dos Santos (2018; 2021) does not give a specific rate of "failed" repetitions among her NNSs. It is thus unclear whether the accurate repetition of SRT utterances sits systematically around 70% for NNSs, or whether my speakers' struggle was unprecedented.

In general, I found that longer utterances and utterances containing obscure vocabulary were the most difficult for participants to repeat. I also observed that utterances with *il(s)* near the beginning yielded the best retention of the pronoun, regardless of the utterance's overall length. To this end, Table 4.1 gives examples of utterances that were uniformly easy and difficult for speakers to repeat:

Table 4.1: "Easy" and "Difficult" utterances[12]

Easy (> 80% *il(s)* retention)	Difficult (<50% *il(s)* retention)
Ouais **i'** bosse tout l(e) temps c'est abuser	Je m'interroge sur la façon donT **i'** conçoivent l'enseignement en général
iL a accepté d'adopter un amendement polémique	Donc sur mon site préféré du coup je vais aller voir les décos qu'**iL**s ont
euh non, **ILs** font la fête dans un club quoi	Quant aux économistes, **iLs** s'inquiètent des thèses sur les bienfaits de la propriété privée
i's ont été poursuivis pour avoir négligé leurs obligations	**i' a** parlé au nom des petits Etats insulaires dont la situation est pire

Individually, participants ranged widely in how many of their tokens could be kept. Participant I had the most usable tokens with 28 of 31 (90.3%), while participant C had the fewest, with only 10 of 31 kept (32.2%). C does seem to be an outlier, however; after C's ten, the next lowest number of tokens kept was E with 19 (61.3%). The number of usable tokens per speaker is displayed in Table 4.2.

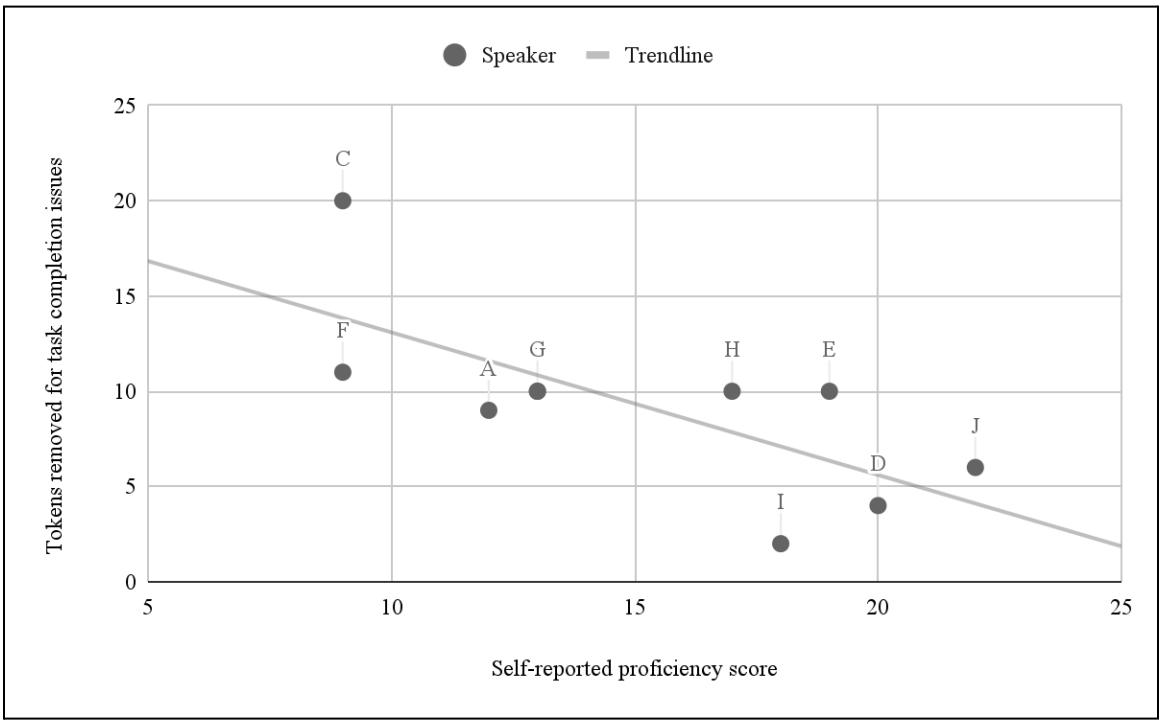
A Pearson's Correlation coefficient was calculated to measure the correlation between number of tokens discarded for task completion issues (i.e.., not those with ambiguous /l/) and BLP proficiency score. A moderate and statistically significant negative correlation was found (r = -0.712, p = 0.021). This finding is also reflected in Figure 4.1. Overall, then, assuming that ability to complete repetition tasks can provide an objective measure of proficiency, I conclude that the self-reported BLP proficiency scores are a good measure of participants' actual language abilities.

[12] English translations of these utterances are given in Appendix D.

Table 4.2: Number of Usable Tokens by Speaker

Participant	Tokens discarded for ambiguous /l/	Tokens discarded for task completion issues	Total tokens kept	BLP proficiency score
A	1	9	21	12
B	0	10	21	13
C	1	20	10	9
D	1	4	26	20
E	2	10	19	19
F	0	11	20	9
G	0	10	21	13
H	0	10	21	17
I	1	2	28	18
J	2	6	23	22

Figure 4.1. Tokens removed by proficiency score



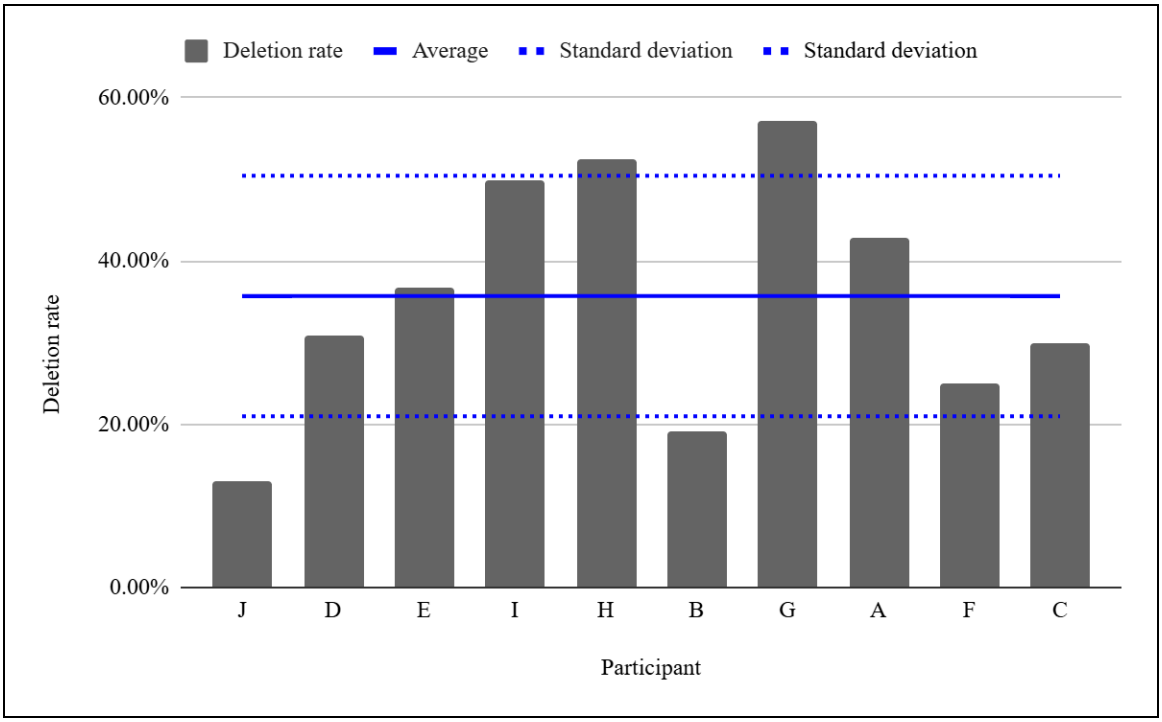
4.2 /l/ Deletion

4.2.1 Findings

As described above, 210 tokens were deemed suitable for analysis. Of these, 134 had /l/ retained and 76 had /l/ deleted for an overall deletion rate of 36.2%.

Among individual speakers, deletion rates ranged from 13% to 57.1% with a median of 30.8%. These results are presented in Figure 4.2; participants are presented from left to right in descending order of BLP proficiency score.

Figure 4.2: Deletion rate by speaker



A Pearson's Correlation coefficient was again calculated to measure the correlation between overall /l/ deletion and self-reported French proficiency. The coefficient was found to be negligible and insignificant (r = -0.057, p = 0.875) indicating that more proficient speakers were not more likely to delete /l/ overall. These findings are represented in Figure 4.3.

However, speaker J appears to be an extreme outlier in this data. Although J had the highest proficiency score at 22, the speaker also had the lowest deletion rate at 13%. Removing J from the data yields a Pearson's Correlation coefficient of r = 0.309 (p = 0.419), which suggests a moderate positive correlation between language proficiency and deletion rate, in line with my hypothesis. However, perhaps due to the small sample size, the correlation was still not significant. The correlation is displayed in Figure 4.4.

Figure 4.3: Deletion rate by proficiency score

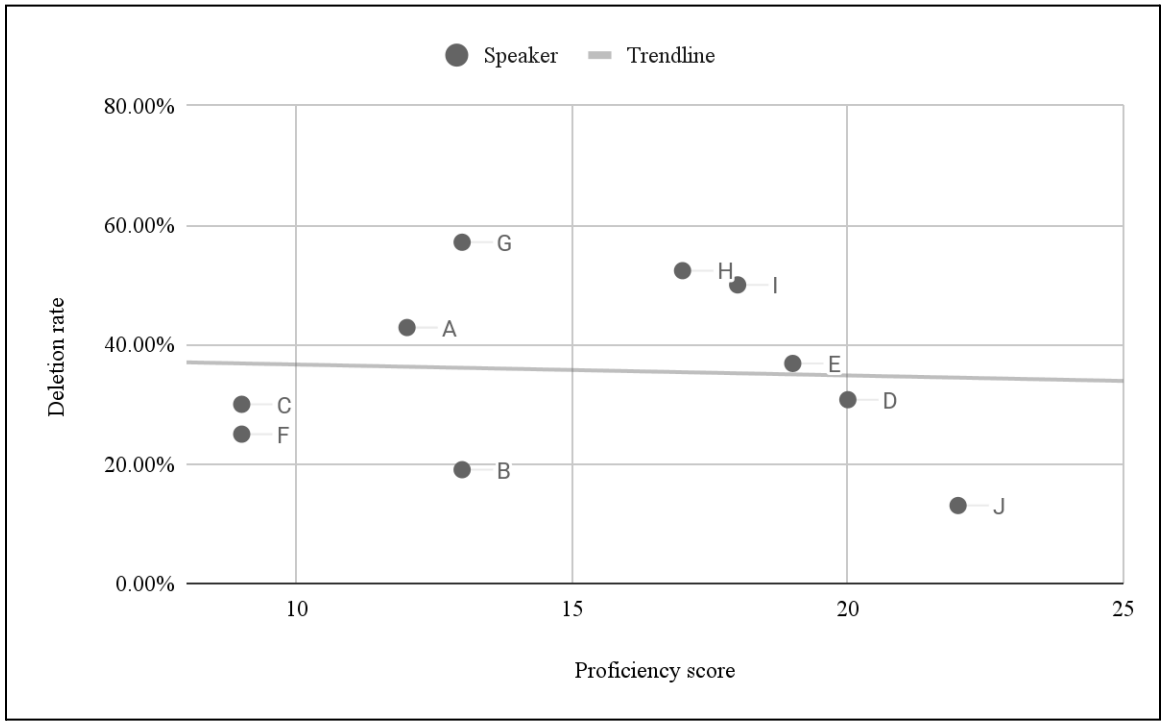
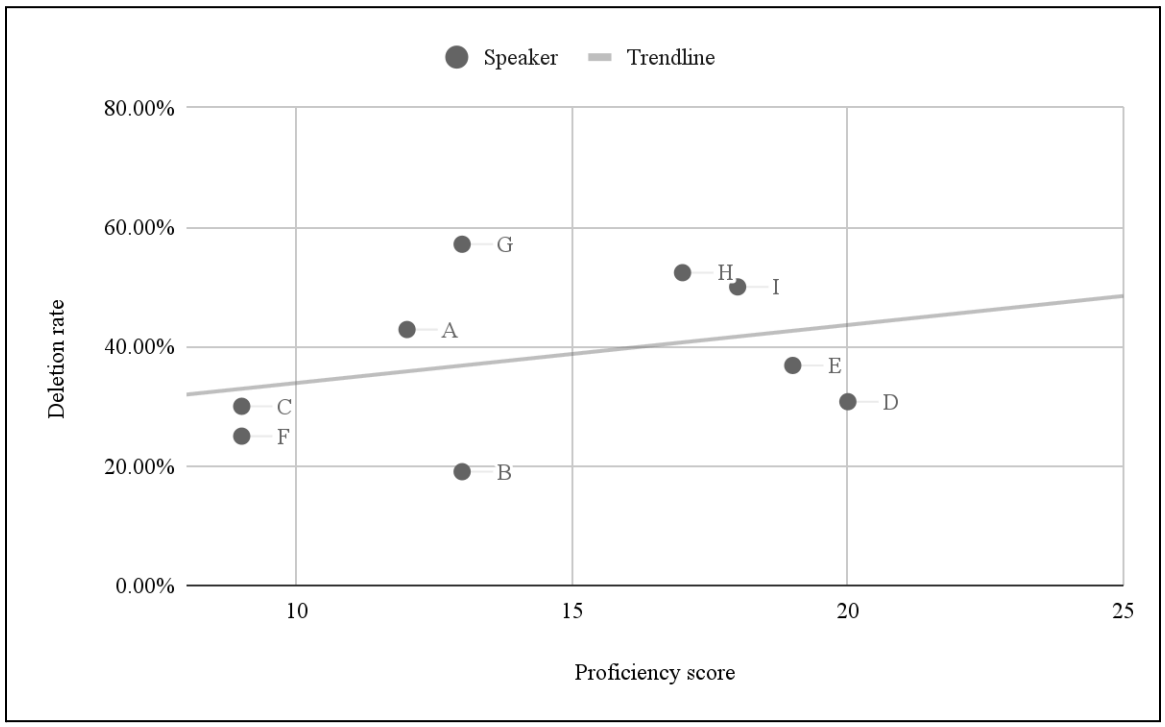


Figure 4.4: Deletion rate by proficiency score (J removed)



In sum, speakers deleted /l/ at a rate slightly above the average reported in the literature. There was no statistically significant demographic predictor for rate of /l/ deletion, however speaker proficiency does show the highest correlation of any variable tested. Of course, however, these findings do not speak to whether /l/ deletion operated in a native-like way. I now turn to a discussion of these results on the group level.

4.3 The effects of linguistic variables

A mixed-effects multiple logistic regression (variable rule) analysis was conducted using Rbrul (Johnson 2009). Variable rules are the preferred method of analysis in sociolinguistic research because they are able to simultaneously assess the effects of multiple factors (i.e., token type and phonological context) on a single linguistic variable (Johnson 2009). As an improvement over its predecessor, VARBRUL, Rbrul is able to account for random effects, such as speaker. Kennedy Terry (2017) notes, “This is especially important for L2 learners who do not, as a group, meet the typical requirements of a “speech community” and who are expected to exhibit high levels of individual variation” (562). “Speech community” is used here to mean a group who share a common dialect. An Rbrul analysis was thus selected as the most appropriate for this data. Speaker was treated as a random effect, /l/ retention was the dependent variable, and independent variable factor groups were: token type, following segment, presence of /l/ in the original token, formality, and homogeneity of the original token. The computed factor weights for each factor group are given in Table 4.3.

Table 4.3: Rbrul factor weights for /l/ deletion by factor group

Factor groups[13]	n	% deletion	Factor weight	Log odds
Token type				
il y a	36	66.7%	0.800	1.385
ils	63	46.0%	0.609	0.442
Impersonal *il*	57	26.3%	0.376	-0.507
Personal *il*	54	14.8%	0.211	-1.320
Following segment				
Il + y a	36	66.7%	0.751	1.105
Il + C	76	34.2%	0.413	-0.352
Il + V	98	26.5%	0.320	-0.753
Original token				
/l/ deleted	91	52.7%	0.666	0.691
/l/ present	119	23.5%	0.334	0.334
Formality				
Formal	91	42.9%	[0.429]	0.282
Informal	119	31.1%	[0.311]	-0.282
Homogeneity				
Homogeneous	54	38.9%	[0.522]	0.0863
Heterogeneous	156	35.3%	[0.478]	-0.0863

Token type was found to be the most significant predictor of /l/ deletion. *Il y a* strongly favored deletion and *ils* slightly favored deletion. Impersonal *il* moderately disfavored deletion, and personal *il* strongly disfavored deletion. Notably here, I chose to treat *il y a* as a separate token type from impersonal *il* on the basis that it favors deletion much more strongly than other contexts for impersonal *il*. However, for the sake of comparison with previous literature, a

[13] Factor groups not selected as statistically significant are reported in square brackets.

second analysis was run with *il y a* as part of the impersonal *il* category. Under this model, the factor weight for *ils* slightly increased to 0.654, the factor weight for impersonal *il* greatly increased to 0.612, and the factor weight for personal *il* slightly increased to 0.251. As predicted, these findings suggest that my speakers are sensitive to native-like constraints on deletion and that the SRT is a viable paradigm for testing such awareness.

Following segment was also found to be highly significant. When treated as a separate category, *il y a* strongly favored deletion (0.751). *Il* + C slightly disfavored deletion (0.413), and *il* + V moderately disfavored deletion (0.320). When *il y a* was incorporated into the *il + C* category, it had a factor weight of 0.606, and *il +V* had a factor weight of 0.394. Again, these findings generally corroborate the hypothesis that NNSs adhere to native constraint orderings, and the effect is generally stronger here than as seen by Kennedy Terry (2017). A comparison of factor weights between the two studies is given in Table 4.4; note that Kennedy Terry's data includes deletion in *elle(s)* in addition to *il(s)*. However, given that Kennedy Terry found only a 3.2% deletion rate for *elle(s),* I take our results to be comparable.

Table 4.4: Comparison of factor weights with Kennedy Terry (2017)

Factor	Factor weight in Kennedy Terry (2017)	Factor weight in present study
Ils	0.8	0.654
Impersonal *il*	0.65	0.612
Personal *il*	0.38	0.251
Il + V	0.55	0.606
Il + C	0.46	0.394

The presence versus absence of /l/ in the experimental utterance was also found to predict /l/ deletion, contrary to my hypothesis. Utterances without /l/ favored deletion (factor weight 0.666), and utterances with /l/ disfavored deletion (factor weight 0.334). This result is perhaps unsurprising given the known influence of accommodation effects on L2 speech (see, for example, Burin (2018)). It seems likely that NSs would also exhibit some degree of accommodation effect, and it would be interesting to test the strength of that effect compared to NNSs. In any case, these findings indicate that a speaker's performance is not entirely dependent on their mental representation of the variable, but is also influenced by the experimental utterance.

Homogeneity and formality were not found to be statistically significant predictors of /l/ deletion. In the case of homogeneity, this is the expected result. The "ideal" speaker's deletion should not be influenced by the homogeneity of the token, and the fact that NNSs' behavior was not influenced by homogeneity is another indicator of the viability of the task. On the other hand, the lack of effect for formality is somewhat puzzling. It is true that Howard et al. (2009) does not find an effect for style with his NNSs, although Sax (2003) does, and Buson et al. (2018) finds that the absence of /l/ in stylistically formal sentences prompted their NSs to "repair" by inserting /l/, suggesting that the variable is indexically informal for NSs. Thus, the most obvious interpretation for my data is that my speakers, while sensitive to linguistic constraints, are not aware of the indexical value of the variable.

Having discussed corpus-level findings, I now turn to my second set of research questions, which concern the performance of individual speakers.

4.4 The Performance of individual speakers

4.4.1 Individual deletion rates

To briefly revisit the findings in section 4.2.1, the statistical significance of individual variation can be confirmed by running an Rbrul model with speaker as a factor group instead of a random effect. The factor weights for each speaker under this model are given in Table 4.5; deletion percentages are repeated for the sake of comparison. This analysis confirms that there is a significant effect for individual variation in the corpus.

Table 4.5: Factor weights for individual speakers' deletion

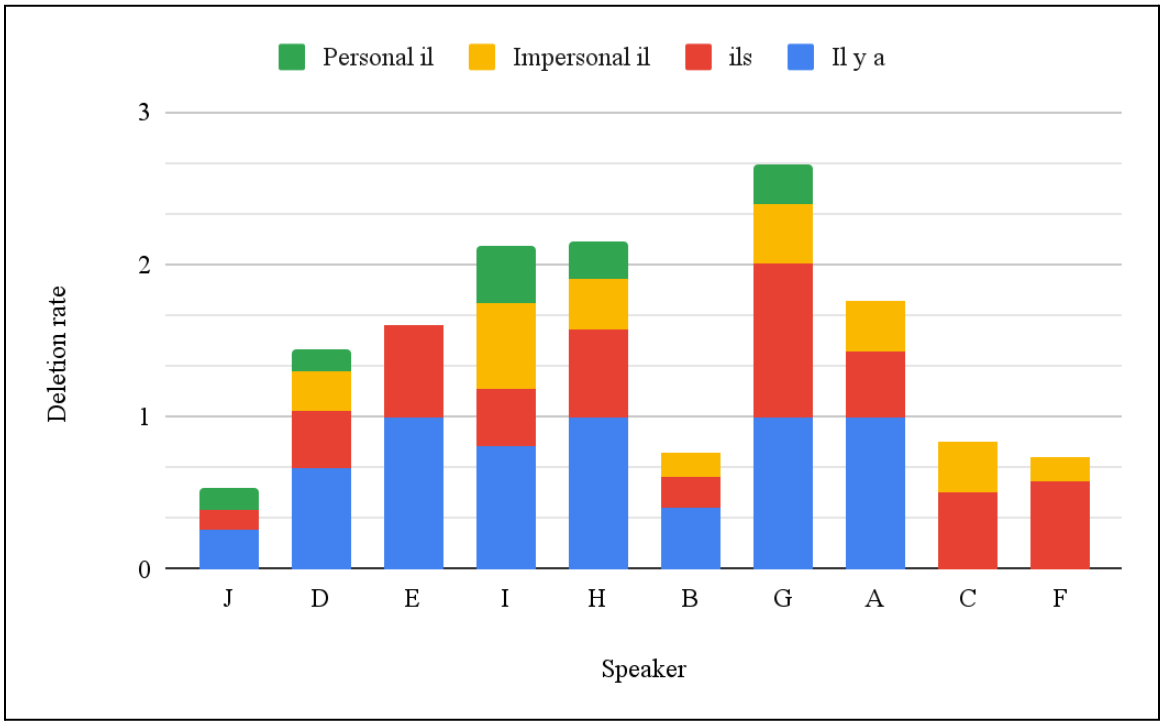
Participant	Factor weight	Deletion %
G	0.720	57.1
H	0.679	52.4
I	0.658	50
A	0.591	42.9
E	0.529	36.8
D	0.461	30.8
C	0.452	30
F	0.391	25
B	0.312	19
J	0.224	13

4.4.2 Individual deletion by token type

For each speaker, the rate of deletion for each token type was calculated. These rates are visualized in Figure 4.4; I use deletion rates instead of factor weights in this section because the small number of tokens per group resulted in many factor weights of zero or one, obscuring the

possibility for more nuanced inter-speaker comparison. Note that a bar height of one indicates 100% deletion; for example, speakers A, E, G, and H deleted /l/ in 100% of their *il y a* tokens. A taller stacked bar height thus represents higher overall deletion. Based on the native speaker norm, we would expect to see *il y a* (blue) be the tallest segment. *Ils* (red) should be the next tallest, followed by impersonal *il* (yellow), then personal *il* (green). Speakers are presented from left to right in descending order of BLP proficiency.

Figure 4.4: Individual deletion rates by token type



An in-depth analysis of these results will be reserved for the discussion. Here, it suffices to observe that there is indeed a degree of individual variation. Broadly, speakers A, B, D, G, and H all exhibited the expected pattern of deletion in *il y a* >> *ils* >> impersonal *il* >> personal *il.*

While several of these speakers did have a high overall deletion rate, they are not uniformly the highest deleters in the corpus, nor do they have the highest proficiency scores.

Speakers C, F, and J do seem to be cases where a low overall deletion rate correlates with non-adherence to native-like patterns, though their deletion does not appear to be completely random. C and F both display the expected *ils* >> impersonal *il* >> personal *il* pattern and simply do not delete at all in *il y a;* J, the lowest deleter in the corpus, did at least favor *il y a* for deletion. Notably, C and F are also the speakers with the lowest BLP proficiency scores; J has already been established as an outlier. Speakers E and I thus emerge as the only two highly-deleting speakers who do not adhere to native-like patterns. Even they, however, respect certain heuristics: *il y a* is deleted at the highest rate, and personal *il* is deleted at the lowest rate. The overall narrative thus seems to be: a low deletion rate precludes adherence to native-like patterns; a high deletion rate is often, but not always, associated with native-like adherence.

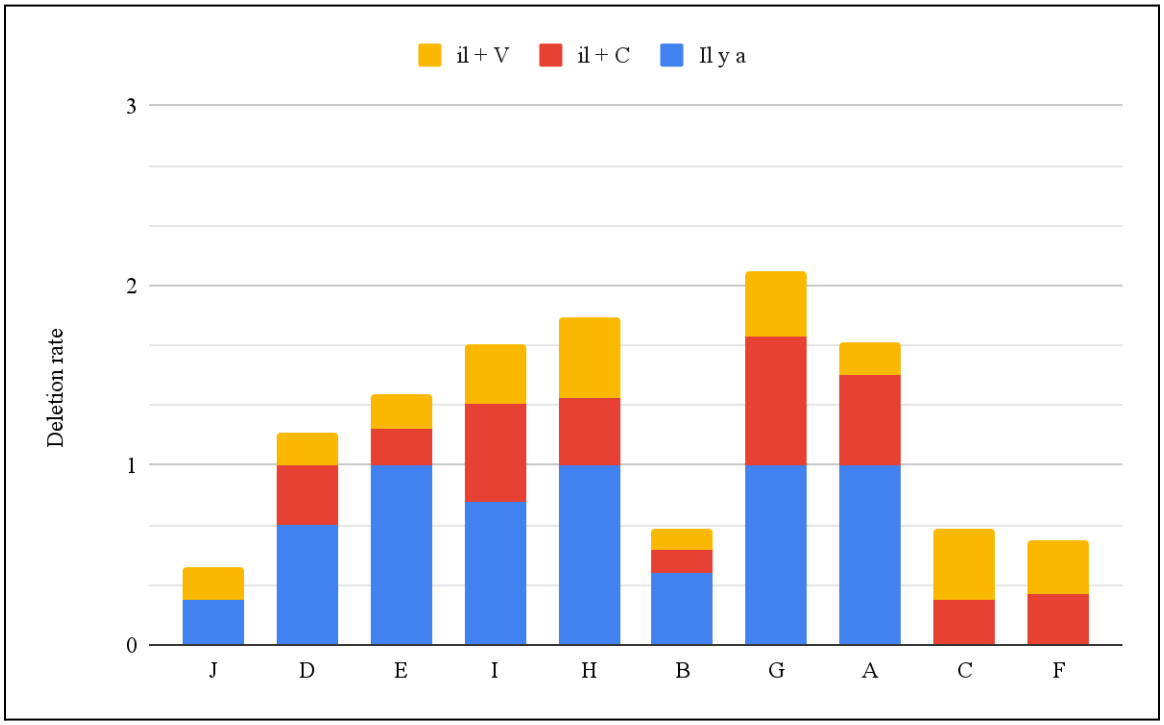
4.4.3 Individual deletion by phonological context

In addition to token type, each speaker's rate of deletion was calculated for each of the three phonological contexts. These findings are presented in Figure 4.5, to be interpreted in the same way as Figure 4.4. Given the native speaker norm, we would expect deletion in *il y a* (blue) to be very high, followed by *il* + C (red), and the rate of deletion for *il* + V (yellow) should be very low.

Four speakers followed the expected pattern of *il y a* >> *il* + C >> *il* + V: I, D, G, and A. Speakers B and E also approximated the pattern but had equivalent deletion rates for *il* + C and *il* + V. As with above, H is an outlier as a highly-deleting speaker who does not adhere to the NS pattern; H deleted in *il* + V more than *il* + C. Otherwise, the three speakers who deviate completely from the the NS pattern are the three lowest deleters: J, C, and F. All three delete in

il + V more than *il* + C, which is highly unexpected given the near-categorical NS deletion for *il* + V. Here again, then, a low overall deletion rate seems to preclude any kind of adherence to native-like phonological constraints; it is less clear whether higher proficiency predicts adherence or not.

Figure 4.5: Individual deletion rates by phonological context

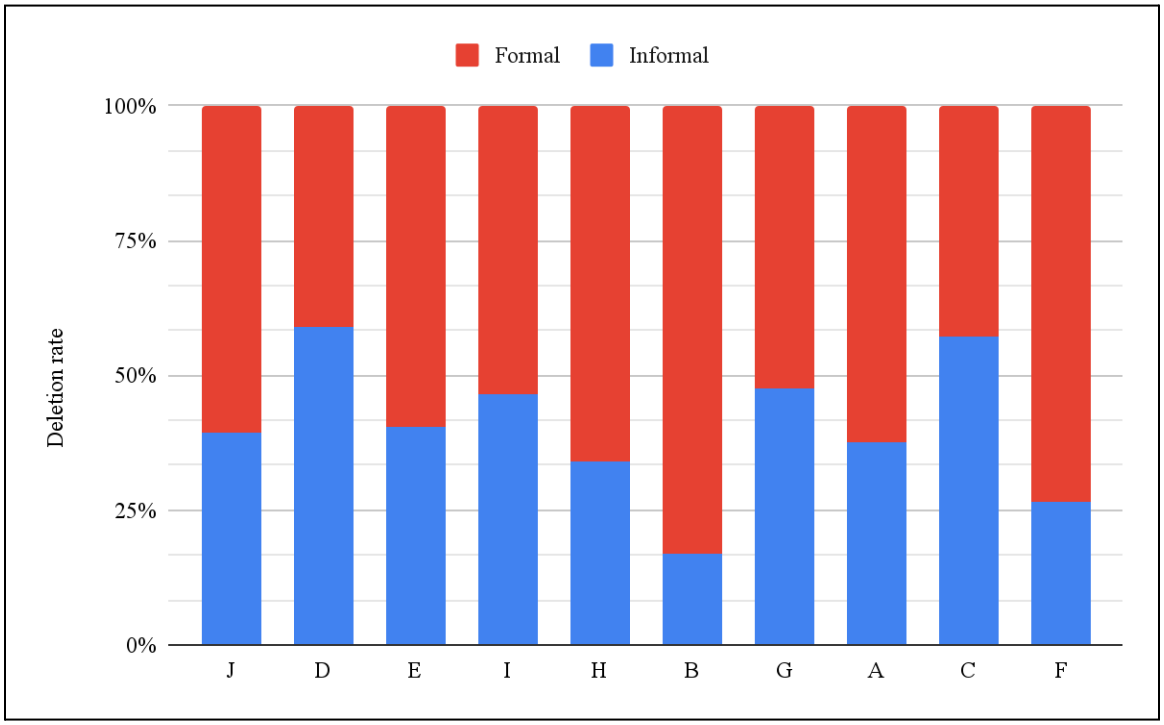


4.4.4 Individual deletion by formality

Finally, individual speakers' rate of deletion for the formal and informal contexts was calculated. These results are given in Figure 4.6; we would expect a higher rate of deletion in informal contexts (blue) than formal contexts (red) based on the NS norm. Given that a significant effect for formality was not found on the corpus level, we might wonder whether any speakers at all show this sensitivity, or whether even the most sociolinguistically proficient speakers do not.

Note here that Figure 4.6 should be interpreted slightly differently from Figure 4.4 and Figure 4.5. Because I am only comparing two variables, the overall bar height represents 100% of the speaker's tokens in which deletion occurred, and the red and blue areas represent the percentage of overall deletion that each category accounts for.

Figure 4.6: Percentage of deletion for formal and informal categories by speaker



In fact, every speaker besides D and C deleted more in the formal context than informal; D and C were neither the most proficient nor highly-deleting speakers. An additional four speakers had less than a 15% difference between the categories. Speakers F, A, B, and H were the only ones to clearly prefer one category over another, and they all preferred the "wrong" category, according to the NS norm. It is thus clear that a lack of sensitivity to formality constraints is not due to a lack of overall competence around the variable, but a more widespread

issue in L2 speech. In the following section, I analyze these results further in the broader context of L2 sociophonetic competence.

5. Discussion

This project set out to address two main gaps in the literature around L2 French /l/ deletion specifically, and around L2 sociolinguistic competence more generally. First, can the Sociolinguistic Repetition task (SRT) be effectively applied to test for NNSs' awareness of linguistic constraints on sociolinguistic variables, and if so, what new insights does this methodology offer over sociolinguistic interviews? And second, at what point in the development of sociolinguistic competence is awareness of linguistic constraints acquired? I discuss my findings around each of these two questions in turn.

Overall, I did find the SRT to be an effective tool for assessing NNSs' knowledge of their L2's variable grammar. My participants deleted /l/ at a rate consistent with the literature, more proficient speakers tended to delete /l/ at a higher rate, and all of my speakers exhibited at least some /l/ deletion. Likewise, the SRT yielded enough tokens for significant corpus-level Rbrul results, and the analysis produced factor weights in line with NS norms. All of these facts suggest that the SRT can effectively be adapted to assess participants' mental schemas around a single variable. That said, I did not find significantly higher deletion rates or factor weights than previous L2 /l/ deletion studies, as I hoped I would. Deletion rates reported in the literature for speakers of the same demographic have been 22.7%, 62%, 36%, and 20.5%. My rate of 36.2% falls at the upper end of this range if we take Sax (2003)'s 62% to be an outlier, but it certainly does not represent a remarkably higher rate. Likewise, while my Rbrul factor weights show a stronger effect for phonological context than has been reported in the literature, my factor weights for token type are roughly equivalent to those found by Kennedy Terry (2017).

I see three possible explanations to account for these results. First, perhaps the gap between L2 competence (i.e., this task) and performance (i.e., sociolinguistic interviews) is not as significant as predicted. This would be somewhat surprising given the markedly different deletion rates reported in French & Beaulieu's (2016) planned and spontaneous language tasks, but it does not seem completely out of the realm of possibility, especially given that French & Beaulieu's participants had a much lower proficiency than the participants in the present study. This finding would mean that sociolinguistic interviews are indeed a viable method of testing NNSs' competence, which would be a valuable finding in and of itself. To test this hypothesis and to fully validate the sociolinguistic interview's place in SLA methodology, it would be interesting to conduct both the SRT and sociolinguistic interviews with the same participants and compare /l/ deletion patterns.

As a second possible interpretation, perhaps my speakers were simply less proficient than those in previous studies. This seems anecdotally improbable; given my personal connections to many of the speakers, I can confirm that they speak French at a very high level for classroom learners. However, it is true that the participants in many past studies had spent an entire year abroad, whereas most of my participants had only spent 10-15 weeks (speakers E and I are the exceptions). Given Kennedy Terry (2017)'s finding that the strength of NNSs' social connections to NSs while abroad significantly predicts /l/ deletion even among uniformly advanced speakers, perhaps this limited time abroad limits the /l/ deletion in my corpus.

Finally, I find the most plausible explanation to be that the SRT was simply too hard for my participants. The 32% incomplete repetition rate attests to this difficulty, and even among usable tokens, my participants repeated very few utterances word-for-word. Likewise, almost all participants commented on the difficulty of the task after completing the experiment. It thus

seems likely that the cognitive effort required to repeat my SRT utterances effectively turned it into a performance-based task. In future iterations of the experiment, it would be prudent to reduce the length of experimental utterances as much as possible and to include vocabulary that is familiar to NNSs.

To this end, utterances' homogeneity was not, in fact, relevant to my research questions, nor was it significant to my results. In order to achieve more pertinent results, an improved version of this SRT would not include any markedly formal/informal vocabulary which would be unfamiliar to NNSs. Instead, (non)homogeneity would be created by having /l/ deletion occur in sentences where it is strongly favored (i.e., *ils* + C) and strongly disfavored (i.e., personal *il* + V). In this hypothetical version of the SRT, an effect for token homogeneity would be indicative of better sociolinguistic competence, and I would expect to see much stronger effects for different factor weights.

Overall, however, the version of the SRT used in this experiment did seem to test NNSs' sociolinguistic competence just as well as other experimental paradigms. This is a triumph given the SRT's streamlined data collection process; experimental sessions are relatively short, and, crucially, analysis does not require transcription. In the present study, the analysis of each speaker's 31 utterances took approximately 45 minutes, meaning that the SRT would be easily scalable to large corpora. I thus take the viability of the SRT as an alternative to sociolinguistic interviews as one of the most valuable findings of this research.

As for my second strand of research questions, I found a high level of individual variation for adherence to native-like factor weights. However, given the nature of my data, it was difficult to ascertain if there was any correlation between deletion patterns and overall deletion rate. A statistical metric to ascertain such a correlation will be an important subject of future research. In

the meantime, a discussion of a few individual speakers' performances as case studies will be the most telling.

The most highly-deleting speaker in my corpus was G, with an impressive deletion rate of 57.1%. G's self rated proficiency score was below average (they rated their language abilities as a three out of six for speaking, reading, and understanding, and as a four for writing), and their language attitude and use scores were among the lowest in the corpus. Their number of tokens removed for task completion issues was likewise slightly above average, suggesting below-average proficiency. It is worth noting here, however, that other factors may be at play, such as attention and memory. In terms of language background, G reported learning French in a classroom setting and having eight year's worth of classroom French. In terms of token type, G deleted in 100% of *il y a,* 100% of *ils,* 40% of impersonal *il* and 25% of personal *il,* which shows excellent sensitivity to NS norms. Likewise, for phonological context, G deleted in 71.4% of *il* + C and 36.8% of *il* + V. G thus seems to exemplify a speaker who, despite relatively low proficiency, has completely acquired variable /l/ deletion.

G makes an interesting contrast with speaker D, another speaker who well-approximated NS variable rules. D had one of the highest proficiencies in the corpus based both on BLP score and number of usable tokens. However, D's deletion rate was 30.8%, below both the average and the median rate. D's deletion by token type was 66.7% for *il y a,* 37.5% for *ils,* 25 % for impersonal *il,* and 14.3% for personal *il,* approximating NS norms. D likewise deleted in 33.3% of *il* + C and 18.2% of *il* + V, and D was one of the only speakers to delete more in informal contexts than formal. Thus, despite a much lower overall deletion rate, D seems to have mastered the native-like rules as well as speaker G. Clearly, use rate alone cannot paint a full portrait of that variable's acquisitional status.

To this end, the deletion of speaker C makes for another telling case. C had the lowest BLP proficiency scores and lowest numbers of usable tokens. Their deletion rate was 30%, similar to D's. However, their deletion patterns were distinctly non-native-like: deleting more in *ils* and impersonal *il* than *il y a* and more in *il* + V than *il* + C. It is thus clear that C's 30% deletion rate in fact reflects much lower sociolinguistic competence than D's 30.8% deletion.

Finally, speaker J's behavior is intriguing. Speaker J learned French in an immersion classroom and had the highest BLP proficiency and third-highest number of usable tokens. However, they had the lowest deletion rate (13%) and clearly did not abide by native-like factor weights, deleting more in personal *il* than impersonal *il* and more in *il* + V than *il* + C. Like Nadasi et al. (2005)'s immersion speakers who deleted /l/ at only 2%, the immersion context seems to be uniquely inhibitory to the development of /l/ deletion.

In sum, from these case studies, three "levels" of sociolinguistic competence emerge:

1. Speakers such as J with very low deletion rates and low adherence to the native variable grammar (irrespective of overall French proficiency)
2. Speakers such as C with an average deletion rate and low adherence to the native variable grammar
3. Speakers such as D and G with average to high deletion rates and good adherence to the native grammar

From this classification, the following narrative can be constructed: a speaker can acquire the usage of a variable to an extent without also acquiring its patterns of use, but after a certain threshold, variable use rate and patterns of use develop together. No speaker uses the variable at a very high rate but with markedly incorrect patterns of use. This narrative, and especially the existence of speakers such as C who delete "incorrectly" but at an average rate, strongly suggests

the possibility that the acquisition of rate and patterns of use should be treated as potentially independent processes in future research. Likewise, contrasts such as those between speakers D and G absolutely negate the validity of variable usage rate as an objective marker of sociolinguistic competence. Even without a definitive conclusion, the mere finding that native-like usage rates do not perfectly correlate with native-like usage patterns opens new and valuable avenues for research at the intersection of SLA and variationist sociolinguistics.

Speakers D and in J in particular raise interesting questions for future research. From D's low deletion rate but high adherence to NS rules, I speculate: perhaps D has fully acquired the variable grammar and their low /l/ deletion rate is motivated by a stylistic choice instead of a lack of competence. If this were the case, it would be an even stronger motivation for examining individual adherence to native-like factor weights in L2 sociolinguistic competence studies. As for speaker J, the question of why immersion settings seem especially non-conducive to the acquisition of /l/ deletion certainly merits further research. It is especially interesting to wonder whether J and Nadasi et al. (2005)'s immersion speakers were themselves taught by non-native speakers of French and whether extensive input from non-native speakers might somehow ossify L2 speech. If J and Nadasi et al.'s speakers were taught by NSs, one might wonder instead whether immersion schools are particularly liable to reinforcing prescriptive norms.

Next, the lack of effect for formality on /l/ deletion in my corpus brings several interesting questions to light. One interpretation of these results would be that all of my speakers – even those who conformed to the native grammar in every other way – were simply not sensitive to the effect of formality. If this finding holds in future research, it would seem to suggest that linguistic constraints on variable use are somehow easier to acquire than stylistic

constraints. This discrepancy could perhaps be explained by an appeal to universal grammar, if we take linguistic constraints to be reflective of broader cross-linguistic patterns.

A second possible interpretation is that the tokens in my SRT were not distinct enough for my participants to identify them as formal or informal. This seems improbable given that many tokens were taken directly from Buson et al. (2018) who did validate the (in)formality of their tokens in an experiment separate from the SRT. Perhaps, though, the issue rests more on the speakers than the tokens; not that speakers were insensitive to a formality effect, but that they were not proficient enough to assign a formality index to the tokens in the first place, or else that they were listening simply to memorize and repeat and were not paying attention to the semantic content of the utterances.

Finally, and perhaps most probably, it is possible that /l/ deletion is not as strongly associated with informality as the (limited) literature has put forth. Indeed, this possibility is indicative of a much larger issue in the present study, and in French variationist SLA work in general: that it is very difficult to establish a definitive native speaker norm. What this study has taken to be the "gold standard" of /l/ deletion is in fact representative of only two relatively small corpora, each collected over thirty years ago. To definitively understand the nature of sociolinguistic competence in L2 French, more variationist work must first be done on the L1 variety.

As a final point, although a formal acoustic analysis was not conducted as part of this study, the fact that many tokens were so difficult to initially classify as "/l/ retained" or "/l/ deleted" suggests the potential interest of such an analysis. It seems uncontroversial to observe that the deletion of /l/ involves more than simply pronouncing *il(s)* as [i]. Anecdotally, in many cases, the [i] of /il/ with deleted /l/ seems to retain some kind of fantom lateralization, perhaps

due to a slight drop in F2. And even when /l/ is pronounced, it is realized with varying degrees of clarity. This observation makes me wonder: are there multiple allophones of /l/, some fully deleted, some partially deleted (perhaps with multiple allophones of partial deletion), and some fully realized? Is the choice of allophone meaningfully conditioned by linguistic and/or sociostylistic factors in L1 speech? And if so, how good are NNSs at acquiring native-like *phonetic* use of the variable?

There is indeed some precedence for these questions in the existing L2 sociophonetic literature. For example, Dalola & Birdwell (2020) study phrase-final fricative epenthesis, a French sociophonetic variable, in NS and NNSs. They find that, while NSs use several allophones of the fricative depending on context, NNSs generally realize the variable in a single way. From these findings, they conclude:

> This study has instead identified an additional dimension of L2 mastery, namely that of phonetic quality of use. Such a mastery at the level of production would also imply a heightened sensitivity to the perception of these sound variations, affording speakers the ability to decode an additional layer of meaning in an L2. (16)

In my corpus, it is interesting to note that there is a strong and significant correlation between the percentage of a speaker's /l/s initially deemed ambiguous and proficiency score ($r = 0.667$, $p = 0.035$). However, there is no correlation between ambiguous /l/ and deletion rate ($r = -0.134$, $p = 0.712$). With participant J removed the correlation rises to moderate but stays insignificant ($r = 0.354$, $p = 0.349$). These results suggest that production of a "partial deletion" allophone develops alongside overall language proficiency, not alongside sociolinguistic competence.

Clearly, much more research in this area is needed to establish the phonetics of deleted /l/ in L1 speech. However, if we do take the ambiguous /l/ to be native-like,[14] these findings would seem to complicate Dalola & Birdwell's notion of a phonetic dimension of sociophonetic competence, raising the question: is a native-like phonetic realization of a variable more closely tied to overall "accent" than sociolinguistic competence?

Furthermore, I have thus far completely ignored the role of perception in this study. However, it is entirely possible that the acoustic realization of /l/ could be separated from how it is perceived, and furthermore that a NS's perception of /l/ could be different from a NNS's perception; perhaps what I as a NNS perceived to be an ambiguous /l/ in *ils* would be judged "clearly deleted" by a NS given their stronger predictions that /l/ will be deleted in this context.

Such a difference in perception seems entirely plausible, especially given Ishida & Arai's (2016) finding that NS and NNSs have different strategies for phonemic restoration, i.e., reconstructing a sound that has been obscured. Again, however, this discussion remains entirely speculative. Future research will do well to establish a firmer NS phonetic norm and to validate data coding between several researchers. Differences in NS versus NNS' perception of sociophonetic variables will also be interesting to study.

[14] I can corroborate this anecdotally but not quantitatively.

6. Conclusion

This paper set out to investigate the status of /l/ deletion, a sociolinguistic variable, in L2 French. Specifically, the study aimed to fill two gaps in the literature around /l/ deletion. First, is it possible to assess speakers' sociolinguistic competence – that is to say, what they *know* about /l/ deletion – separately from how they actually use the variable in fluent speech? And second, how can we describe and account for individual differences in NNSs' adherence to native-like linguistic constraints on /l/ deletion? Overall, I find that the Sociolinguistic Repetition Task (SRT) can be leveraged to assess NNSs' awareness of linguistic constraints on /l/ deletion. However, it is less clear whether my version of the SRT better assessed competence than previous performance-based studies. I also find that, while my corpus of advanced NNSs generally respected native-like constraints on /l/ deletion, there was indeed a large degree of individual variation, and a speaker's overall deletion rate did not perfectly predict how well the speaker would obey native-like constraints on deletion. This finding suggests that overall usage of a variable should not be taken as a pure indicator of a speaker's sociolinguistic competence in variationist SLA research. It likewise suggests the existence of different stages of sociolinguistic competence: first random usage of the variable, then, once usage reaches a certain threshold, meaningfully constrained usage. Although the existence of such "stages" cannot be definitively confirmed in the present study's small corpus, the findings provide an exciting avenue for further research in the field of variationist SLA.

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Appendix A: Acronyms and abbreviations

Table A.1: Acronyms and abbreviations

Acronym	Meaning
3	Third person
F	Feminine
L1	Native language
L2	Non-native language
M	Masculine
NNS	Non-native speaker
NS	Native speaker
PL	Plural
SG	Singular
SLA	Second language acquisition
%	Grammatical, but less likely to occur

Appendix B: BLP survey questions

Participant ID (assigned by researcher)

Highest level of formal education

At what age did you start learning ENGLISH?

At what age did you start learning FRENCH?

At what age did you start to feel comfortable using ENGLISH?

At what age did you start to feel comfortable using FRENCH?

How many years of classes (grammar, history, math, etc.) have you had in ENGLISH (primary school through university)?

How many years of classes (grammar, history, math, etc.) have you had in FRENCH (primary school through university)?

How many years have you spent in a country/region where ENGLISH is spoken?

How many years have you spent in a country/region where FRENCH is spoken?

How many years have you spent in a family where ENGLISH is spoken?

How many years have you spent in a family where FRENCH is spoken?

How many years have you spent in a work environment where ENGLISH is spoken?

How many years have you spent in a work environment where FRENCH is spoken?

In an average week, what percentage of the time do you use ENGLISH with friends?

In an average week, what percentage of the time do you use FRENCH with friends?

In an average week, what percentage of the time do you use OTHER LANGUAGES with friends?

In an average week, what percentage of the time do you use ENGLISH with family?

In an average week, what percentage of the time do you use FRENCH with family?

In an average week, what percentage of the time do you use OTHER LANGUAGES with family?

In an average week, what percentage of the time do you use ENGLISH at school/work?

In an average week, what percentage of the time do you use FRENCH at school/work?

In an average week, what percentage of the time do you use OTHER LANGUAGES at school/work?

When you talk to yourself, how often do you talk to yourself in ENGLISH?

When you talk to yourself, how often do you talk to yourself in FRENCH?

When you talk to yourself, how often do you talk to yourself in OTHER LANGUAGES?

When you count, how often do you count in ENGLISH?

When you count, how often do you count in FRENCH?

When you count, how often do you count in OTHER LANGUAGES?

Are you currently studying or living abroad in a Francophone country?

How well do you speak ENGLISH? (six-point likert scale)

How well do you speak FRENCH? (six-point likert scale)

How well do you understand ENGLISH? (six-point likert scale)

How well do you understand FRENCH? (six-point likert scale)

How well do you read ENGLISH? (six-point likert scale)

How well do you read FRENCH? (six-point likert scale)

How well do you write ENGLISH? (six-point likert scale)

How well do you write FRENCH? (six-point likert scale)

I feel like myself when I speak ENGLISH. (six-point likert scale)

I feel like myself when I speak FRENCH. (six-point likert scale)

I identify with an ENGLISH-speaking culture. (six-point likert scale)

I identify with a FRENCH-speaking culture. (six-point likert scale)

It is important to me to use (or eventually use) ENGLISH like a native speaker. (six-point likert scale)

It is important to me to use (or eventually use) FRENCH like a native speaker. (six-point likert scale)

I feel capable of speaking in a native-like way in ENGLISH. (six-point likert scale)

I feel capable of speaking in a native-like way in FRENCH. (six-point likert scale)

I want others to think I am a native speaker of ENGLISH. (six-point likert scale)

I want others to think I am a native speaker of FRENCH. (six-point likert scale)

Appendix C: Full BLP results

Table C.1: Detailed language proficiency

Participant	How well do you…				Composite Proficiency Score (0-24)
	Speak French? (0-6)	Understand French? (0-6)	Read French? (0-6)	Write French? (0-6)	
A	3	3	3	3	12
B	4	4	3	2	13
C	3	2	2	2	9
D	5	N/A[15]	5	5	20
E	4	5	5	5	19
F	2	3	3	1	9
G	3	3	4	3	13
H	4	4	5	4	17
I	5	5	4	4	18
J	6	6	6	4	22
Mean	3.9	3.5	4	3.3	14.7

[15] The participant did not respond to this question on the BLP. A score of 5 was inserted for the sake of a representative composite score.

Table C.2: Detailed language attitude scores

Participant	I feel capable of speaking French in a native-like way (0-6)	It is important to me to use (or eventually use) French like a native speaker (0-6)	I want others to think I am a native speaker of French (0-6)	Composite attitudes score (0-18)
A	2	5	4	11
B	4	6	6	16
C	2	6	3	11
D	4	6	5	15
E	4	6	4	14
F	0	5	3	8
G	1	5	0	6
H	4	5	4	13
I	6	6	6	18
J	5	4	5	14
Mean	3.2	5.4	4.0	12.6

Table C.3: Detailed language use scores

Participant	In an average week, what percentage of the time do you use French with friends?	In an average week, what percentage of the time do you use French with family?	In an average week, what percentage of the time do you use French at school/work?	When you talk to yourself, how often do you talk to yourself in French?	When you count, how often do you count in French?	Composite use score (0-50)
A	0%	0%	10%	0%	0%	1
B	0%	0%	10%	10%	0%	2
C	0%	0%	30%	10%	10%	5
D	0%	0%	0%	0%	10%	1
E	0%	0%	10%	50%	0%	6
F	0%	0%	0%	10%	0%	1
G	0%	0%	20%	0%	0%	2
H	10%	0%	10%	10%	10%	4
I	10%	30%	20%	30%	40%	13
J	10%	0%	40%	20%	40%	11
Mean	3%	3%	15%	14%	11%	4.6

Appendix D: Complete list of SRT tokens with English translations

i' y a une quinzaine de jours, le groupe a mis fin à des poursuites civiles.	Fifteen days ago, the group put an end to civil proceedings.
i' est toujours provisoire et sujet à être remis en question.	It is still provisional and open to question
Pour subvenir à ses besoins, i' faut d'autres sources de revenus.	To support himself, he needs other sources of income.
i' a parlé au nom des petits Etats insulaires dont la situation est pire	He spoke on behalf of small island states whose situation is worse
C'esT un pays accueillant où i' fera certes bon vivre.	It's a welcoming country where he will certainly live well
i's ont été poursuivis pour avoir négligé leurs obligations	They were prosecuted for neglecting their obligations
Je m'interroge sur la façon donT i' conçoivent l'enseignement en général	I wonder about the way they see education in general
C'esT un prix symbolique même si le gouvernement___veut pas parler de gratuité.	The cost is symbolic, even if the government doesn't want to refer to it as free
L'écueil majeur concerne la manière qu'on enseigne l'informatique à l'école.	The major stumbling block concerns the way computer science is taught in schools
Certes les débordements dans l'hémicycle étaient____inévitables.	Certainly the outbursts the auditorium were unavoidable
Nous pouvons sans doute nous y soustraire hein.	We can probably get out of it, can't we?
Oh mais aujourd'hui les gens ne se marient plus à 25 ans.	Oh, but nowadays people don't get married at 25 anymore
Quelle que soit la discipline, la connaissance scientifique est___un objectif primoridal	Whatever the discipline, scientific knowledge is a primordial objective
Ce texte___est pas en adéquation avec la volonté du gouvernement	This text is not line with the government's wishes
Marseille, deuxième ville de France, connaît régulièrement des grèves	Marseille, the second largest city in France, regularly experiences strikes

Ben on va se faire un p'tit brunch quoi	Well, let’s go get a little brunch!
Ouais i' bosse tout l(e) temps c’est abuser.	It’s an apartment with a friend, there’s no TV
oh la, iL est déjà 17 heures ? Bon, je me casse	Oh no, it’s already 5 pm? Well, I’m off
les gars j'avais pas vu qu'iL faisait beau comme ça aujourd'hui	Guys, I didn't see it was this sunny today
ben i'est ouf du coup je le vois pas	Well, he’s crazy, in any case I don’t see him
Oui bon en même temps comment iL galère avec sa meuf.	Yes, well, at the same time, how he's having such a hard time with his girl.
Alors iLs ont regardé le truc et iLs ont dit oh mais c’est quoi ça ?!	So they looked at the thing and they were like, what's this?!
euh non, ce___sont pas des danseurs. Ils font la fête dans un club quoi	Uh, no, they’re not dancers. They’re just partying at a club
Essaye de pas m(e) refiler l(e) matos pourri dont t(u) as parlé l’aut(re) fois.	Try not to give me that crap you were talking about last time.
Euh non en vrai j’avais bien les boules de pas capter ses questions.	Well, no, really, I was a bit pissed off that I couldn't understand her questions
Mais si j'ai une gosse avec moi oh la la non je n'arrive pas	But if I've got a kid with me, oh la la, no, I can't do it!
Nan, mais ce n'est pas possible je ne veux pas entendre tout ça hein	No, it's not possible, I don't want to hear all that.
Enfin si, bien sûr. Mais euh, j’y suis jamaiS allée quoi	Of course I do. But uh, I've never been there.
Ce N'était que du mytho franch(e)ment je N'ai jamais fais ça d(e) ma vie	It was just a fib, honestly, I've never done that in my life
C’est quand même un peu chelou ce que je t’ai parlé tout à l’heure	What I told you earlier is still a bit weird
iL y a une absence de pilotage des ressources humaines	There is a lack of human resources management
Oui bon, IL est vrai que les prix Nobel tendent à récompenser des travaux d'avant-garde	Yes, it's true that Nobel prizes tend to reward groundbreaking work

IL ne s’agit que d’une première épreuve pour le budget de l’Etat	This is only a first test for the state budget
iL a accepté d'adopter un amendement polémique	He agreed to adopt a controversial amendment
Etant donné qu’iL travaille à l’hôpital, démissionner esT une décision délicate.	Since he works at the hospital, resigning is a delicate decision.
iLs ont réclamé plus d’investissements publics, dont un milliard d'euros	they called for more public investment, including a billion euros
Quant aux économistes, iLs s’inquiètent des thèses sur les bienfaits de la propriété privée	As for economists, they're worried about theories on the benefits of private property.
Nous ne connaissions même pas le président de cette association.	We didn't even know the president of this association.
De plus, euh non, certains articles évoquaient plutôt une restructuration	And, uh, no, some of the articles were talking more so about restructuring.
Oui bon, la situation est certes éminemment complexe.	Yes, well, the situation is certainly eminently complex.
Les restrictions pétrolières ne constituent qu'un volet essentiel du dispositif	Oil restrictions are only one part of the picture
Euh non, ce que je souhaite, c’est qu’on soit d’arrache-pied sur le dossier	No, what I want is for us to work hard on the dossier.
Les conseils de l'administration ne respectent pas l’ensemble des règles.	The administration's advice doesn’t respect all the rules.
Oui bon, le cadre des négociations esT en cours	Yes well, the framework for negotiations is underway
mais genre c'est ça qu'i' faut se dire parce qu'i y aura pas de fin	but that's what we have to tell ourselves, because there's no end in sight.
bref, i'est là, tout l(e) fric	Anyway, that's where all the money is: between the chic of Monaco and the fame of Saint-Tropez
i' bois du matcha chépa genre depuis trois quatre mois	I’ve been drinking matcha I don’t know for like three four months now
C’est___ un sale gosse j(e) te jure i' est grave.	He’s a brat, I’m telling you, it’s bad

Ouais i' bosse tout l(e) temps c’est abuser.	Yeah, he works all the time, it's too much.
Donc sur mon site préféré du coup je vais aller voir les décos qu'i's ont	So on my favorite site I'm going to check out the decorations they have
Bah tu sais très bien. S’ils dansent, ils veulent être payés comme danseurs	Well, you know very well. If they dance, they want to be paid as dancers.
Tu vas pas mett(re) tes pompes dégueu dans ma caisse, nan ?	You're not going to put your disgusting shoes in my car, are you?
Nan c(e) que j(e) te parlais t(ou)t à l’heure, c’(é)tait un aut(re) plan foireux.	Nah, what I was telling you earlier was just another crappy plan
Écoute t(u) es vachement sympa merci hein.	Listen, you're really nice, thanks.
Oh mais lui pour s’en remett(re) franch(e)ment ça été chaud.	Oh, but it was hard for him to get over it.
Je suis désolée chéri mais tu te sapes mal.	I'm sorry, darling, but you're dressed rather poorly
Mais bref du coup voilà je vais aller me préparer	But anyway, okay, I'm going to get ready.
Ouais ouais, je sais j’suis là, j’vais t’expliquer	Yeah yeah, I know, I'm here, I'll explain